**Design and Implementation of a Smart Laundromat Management System using IoT and Cloud Technologies**

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Dissertation submitted in partial fulfilment of the

requirements for the

Bachelor of Computer Science (Hons)

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CERTIFICATION OF APPROVAL

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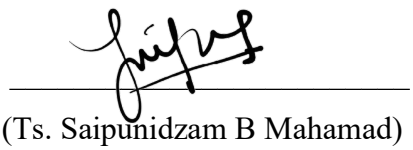
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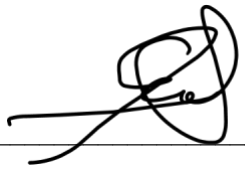


(Ts. Saipunidzam B Mahamad)

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September 2025

## CERTIFICATION OF ORIGINALITY

This is to certify that I am responsible for the work submitted in this project, that the original work is my own except as specified in the references and acknowledgements, and that the original work contained herein have not been undertaken or done by unspecified sources or persons.



(AHMAD LUTFI BIN MOHD ATHZAHA)

## ABSTRACT

This report explores the development of a Smart Laundromat Management System that integrates Internet of Things (IoT) hardware and cloud technologies to improve operational efficiency in public laundry services. Traditional laundromats often suffer from issues such as lack of machine availability visibility, manual booking processes, and limited operational oversight. This project aims to design and implement a cloud-connected IoT prototype that allows users to book machines, monitor machine status in real time, and remotely control machine access through a mobile application. The system architecture incorporates an Arduino-based control unit, a Firebase cloud backend, and a Flutter-based Android application to facilitate user interaction and data synchronization. At this stage, the system is in development, and the expected outcomes include a functional and user-friendly prototype capable of reducing wait times, enhancing machine utilization, and providing owners with actionable insights on usage patterns. The system is designed to operate with low latency and high reliability within a simulated test environment. It is anticipated that the final prototype will validate the practical benefits of combining mobile, cloud, and IoT technologies for small-scale laundromat operations, laying the groundwork for further refinement and deployment in future phases.

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The completion of this Final Year Project marks an important milestone in my academic journey, and I am deeply grateful to all who have supported me throughout this process. This project has given me the opportunity to apply what I have learned in the classroom to a real implementation, and the experience has strengthened my understanding in ways that lectures alone could never achieve. The challenges faced, the skills developed and the knowledge gained have shaped me both academically and personally.

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# CHAPTER 1

## INTRODUCTION

### 1.1 Background of Study

The demand for automated service systems has grown significantly with the advancement of Internet of Things (IoT) and cloud computing technologies. In particular, service-oriented businesses such as laundromats have experienced pressure to modernize their operations in order to meet the expectations of increasingly tech-savvy users. Traditional laundromats often rely on manual processes such as walk-in availability checks, coin-based machine activation, and absence of real-time monitoring. These limitations cause inefficiencies such as long waiting times, poor machine utilization, and customer dissatisfaction. As urbanization increases and more people rely on public laundry services due to space constraints in residential units, the need for a smart laundromat system that offers automated scheduling, machine monitoring, and digital interaction becomes more urgent.

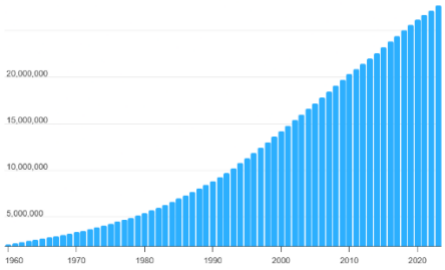



*Figure 1.1 Populations Trends in Malaysia*

Recent developments in IoT enable the deployment of low-cost sensors and microcontrollers that can monitor appliance status, usage time, and energy consumption. At the same time, cloud platforms offer a scalable and centralized approach to manage user data, system logs, and machine operations from any location. When these two technologies are integrated, they form a foundation for smart environments capable of real-time control and decision-making. In the context of laundromats, this integration can lead to systems that allow users to check

machine availability remotely, reserve slots in advance, receive notifications, and monitor washing progress using mobile or web-based applications.

This project aims to design and implement a smart laundromat management system that leverages IoT and cloud computing to optimize machine usage, improve customer experience, and support operational transparency. The system architecture includes microcontroller-based control units attached to laundry machines, connected via Wi-Fi to a cloud database that stores status updates and booking data. The front end consists of a user-friendly interface accessible through a mobile app or web portal, allowing users to interact with the system in real time.

The research also addresses the limitations of existing systems which often lack integration, real-time status updates, and dynamic scheduling features. By developing a working prototype and testing its usability, responsiveness, and reliability, this study contributes to the body of knowledge on smart service systems and presents a practical solution for modern laundromat operations. Furthermore, it supports ongoing efforts to digitize small and medium enterprises (SMEs) in Malaysia, in line with national digital economy agendas. The implementation of such a system can reduces idle machine time, increase customer satisfaction, and enable data-driven decision-making for business owners.

## 1.2 Problem Statement

Laundromat services in numerous urban and semi-urban regions persist in utilising antiquated systems reliant entirely on manual operations. Customers frequently need to visit the location in person to find out the availability, functionality, or present usage of machines. This absence of transparency results in numerous operational inefficiencies, including prolonged waiting periods, unanticipated queues, and unnecessary journeys. For individuals facing time restrictions or dependent on restricted transportation choices, these challenges might lead to considerable inconvenience and dissatisfaction with the service.

Moreover, the lack of an intelligent booking system leads to ineffective time management for both users and business owners. During peak hours or weekends, clients may endure prolonged wait times or be required to return later, leading to a detrimental user experience. Conversely, during off-peak periods, machines may be inactive, resulting in resource underutilisation. This disparity diminishes operational efficiency and impacts the laundromat's revenue. Business owners are hindered in tracking machine usage patterns and comprehending consumer behaviour due to the lack of real-time data collection and reporting functionalities.

Many current laundromat systems lack integration with modern technology such as the Internet of Things (IoT) or cloud systems. Although some premium service providers have begun using digital payments or status display panels, these solutions are frequently isolated and lack a centralised management system. Consequently, critical tasks such as remote status monitoring, automated usage tracking, predictive maintenance notifications, and client engagement via mobile devices remain predominantly unmet. The expense of adopting commercially available smart technologies poses a significant obstacle for several small and medium-sized laundry operators, particularly in developing areas.

The present landscape reveals a distinct disparity between user expectations and the technological proficiency of conventional launderettes. A cheap, scalable, and user-friendly system is required to rectify these operational deficiencies. A cloud-integrated IoT smart laundromat management system might provide real-time machine status updates, mobile

booking capabilities, usage analytics, and remote-control functionalities. This technology would enhance customer satisfaction by allowing users to schedule their laundry sessions in advance, monitor their cycles remotely, and minimise waiting times. It would offer business owners critical information about usage patterns, machine performance, and service demand, resulting in enhanced decision-making and increased operational efficiency.

This project tackles these significant issues by developing and executing a smart laundry system that automates booking, monitoring, and control tasks through IoT and cloud technologies. The objective is to showcase a functional prototype that is efficient, economical, and suitable for actual laundry settings, particularly for small enterprises pursuing digital transformation.

## 1.3 Project Objectives

The objectives of the project are:

- To design and develop a cloud-based IoT architecture with a secure locking mechanism for real-time monitoring and control of laundromat machines.
- To implement a mobile application that connects to IoT devices and displays machine availability to users in real time.

## 1.4 Scope of Study

This project focuses on the development of a smart laundromat management system that integrates Internet of Things (IoT) hardware and cloud-based services to enhance the operational efficiency and user experience of a typical self-service laundromat. The scope is limited to the design, development, and testing of a functional prototype that simulates key features found in real-world laundromat environments, such as machine monitoring, remote access, secure locking, and mobile-based interaction.

The IoT component of the system involves the use of a microcontroller such as Arduino or Raspberry Pi connected to washing machine simulators. These devices detect machine status using simple sensors and transmit data wirelessly to a cloud database. The system also includes a locking mechanism that is electronically controlled to ensure that only authorized users with valid bookings can access the machine during their reserved time slots.

The cloud component is responsible for storing and managing data related to user accounts, machine status, bookings, and usage logs. It also provides a communication bridge between the hardware devices and the user-facing mobile application. The scope includes the implementation of basic security measures such as authentication, encrypted communication, and role-based access to ensure safe operation and data integrity.

The mobile application is designed to allow users to register, log in, view available machines in real time, book a time slot, and receive status updates or notifications. It serves as the primary interface for users to interact with the system and is limited to Android platforms for

the purpose of this project.

This study does not include the integration of payment systems, commercial-scale hardware deployment, or maintenance tracking. The system is tested in a simulated environment, not in an operational laundromat, and the performance evaluation focuses on functionality, usability, and communication accuracy rather than large-scale scalability. Network constraints, environmental conditions, and hardware durability are not deeply explored.

Overall, the scope is confined to the core features necessary to demonstrate a proof-of-concept smart laundromat system. It highlights the feasibility of combining IoT and cloud computing to improve machine usage visibility, user convenience, and operational control in small to medium-sized laundromat settings.

# CHAPTER 2

## LITERATURE REVIEW

### 2.1 Internet of Things Integration for Machine Status Detection

The integration of Internet of Things (IoT) technologies in machine status detection has emerged as a crucial solution in industrial automation and service-based environments such as laundromats, manufacturing, and smart facilities. These systems rely on microcontrollers, wireless communication protocols, and embedded sensors to monitor equipment conditions, collect operational data, and enhance real-time visibility for decision-making and maintenance planning.

Zolkapli et al. (2023) demonstrated a wireless IoT platform designed to monitor vibration and temperature in CNC milling machines. The system used sensors connected to a microcontroller (NodeMCU), transmitting data via MQTT protocol to the Node-RED dashboard. Real-time monitoring enabled early fault detection and automatic email alerts, reducing potential equipment damage and downtime. This study highlights the importance of combining sensor-based detection with efficient data communication networks to support predictive maintenance.

Maulana et al. (2024) extended the application of IoT to a smart machinery monitoring system that integrates mobile applications built with Flutter. Their system connected multiple industrial sensors that measuring temperature, load, and vibration to a mobile platform via wireless protocols. The mobile app featured real-time visualization and automated alerts, enabling users to monitor machine conditions and respond quickly to operational anomalies. Testing showed that the system reduced machine downtime and improved asset management, especially in environments where timely intervention is critical.

Sruthi et al. (2024) proposed a comprehensive IoT-based monitoring architecture that uses Arduino and GSM modules to collect and transmit real-time machine data. The system

included sensors for fire, gas, and environmental conditions and was capable of sending alerts via SMS and updating data to cloud platforms. Their model emphasized the interdisciplinary role of IoT in safety management and continuous equipment tracking. This approach also addressed integration challenges, including data security, scalability, and environmental adaptability.

Despite the benefits of IoT integration, challenges persist. These include maintaining stable wireless connectivity, ensuring data privacy, and accommodating the diverse infrastructure needs of different machines. However, all three studies affirm that IoT-based machine status detection systems improve operational efficiency, enable early fault response, and support remote access through scalable and flexible architectures (Zolkapli et al., 2023; Maulana et al., 2024; Sruthi et al., 2024).

In summary, IoT-enabled machine monitoring systems are key enablers of automation and digital transformation. Their effectiveness depends on the proper integration of sensor technologies, secure wireless communication, and user-oriented platforms such as dashboards or mobile apps.

## 2.2 Sensor Integration and Real Time Data Collection

Sensor integration and real-time data collection are increasingly important in the development of Internet of Things (IoT) systems for smart environments. Researchers have focused on how embedded sensing devices, wireless data transmission, and intelligent processing methods can work together to support the continuous monitoring and analysis of machine status, environmental conditions, and system behavior across various applications.

Syafrudin et al. (2018) developed a multi-sensor IoT platform for automotive manufacturing by integrating temperature, humidity, accelerometer, and gyroscope sensors through a Raspberry Pi. The system combined Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and Random Forest algorithms with cloud services such as Apache Kafka and Storm. This setup enabled real-time fault detection with 100% accuracy, offering strong performance in fast-paced industrial settings.

In agriculture, Aygün et al. (2019) used a microcontroller-to-cloud architecture with regression trees to reduce the need for redundant sensors. Their system monitored key variables such as light, soil moisture, and dew point, achieving 92% accuracy in temperature readings while keeping hardware costs low. Valente et al. (2019) proposed a fog-based architecture using microservices and the Chauvenet criterion to reject outliers in environmental data. Their model supported scalable and efficient processing with a predictable latency profile.

Sensor fusion has also been widely adopted in indoor systems. The system achieved up to 99.09% accuracy, with performance improving as more sensors were integrated. Dautov et al. (2019) explored healthcare applications using a hierarchical data fusion approach that combined biometric signals such as heart rate, oxygen saturation, and movement. Their architecture distributed processing across edge and cloud layers, achieving event processing latency between 66 and 5513 milliseconds.

Recent works have emphasized the role of deep learning in enhancing sensor fusion. Gantla et al. (2025) and Varshney et al. (2025) implemented convolutional and recurrent neural

networks to manage multi-modal data in smart city and biomedical contexts. These systems supported anomaly detection and adaptive control through edge-based processing. Lakshmi et al. (2024) reported a high-throughput IoT pipeline capable of handling 5,000 events per second with a latency of just 50 milliseconds and a precision range between 88.7% and 95.2%.

In summary, current literature presents a broad range of integration techniques, from rule-based models to complex neural networks, depending on the domain requirements. Edge and fog computing architectures are widely adopted to meet the demand for low-latency processing. Nonetheless, challenges such as sensor heterogeneity, data quality, and system scalability continue to require further research for reliable and responsive IoT deployments.

## 2.3 Automation and Decision Support through Cloud and Internet of Things Technology

Integrating cloud computing with IoT infrastructures has enabled real-time automation and intelligent decision-making in various smart service environments. These systems utilize cloud storage, sensor-driven triggers, and event-based frameworks to control operations dynamically and support adaptive behaviour across industries such as manufacturing, energy, agriculture, and home automation.

Salhaoui et al. (2019) presented a three-layer architecture that combines unmanned aerial vehicles, fog nodes, and cloud platforms for real-time control in a concrete plant. The system leverages Node-RED and deep learning for image analysis and anomaly detection, achieving automation with a round-trip latency of 3 seconds. A similar architecture is described by Lam et al. (2022), who used a decentralized local cloud with the Arrowhead framework and semantic technologies for adaptive decision-making in chemical production. Their system demonstrated under-3-second response times for reasoning and policy-driven adaptation, emphasizing its reliability and responsiveness.

The use of standardized frameworks was also evident in Quality of Service control service with a local cloud infrastructure to automate climate systems. The system achieved sub-millisecond latencies and demonstrated significant time savings in engineering tasks. In contrast, Dai et al. (2018) applied fault tree analysis within a cloud-based architecture to create a self-healing water heating automation system. Although performance metrics were not reported, the system was designed for autonomous recovery using IEC 61499-compliant software agents.

Decision support in energy systems was explored by Rosero et al. (2023), who developed a cloud-based platform with machine learning algorithms to forecast load behaviour in a microgrid cluster. The study emphasized autonomous control and adaptive energy management, though quantitative metrics were absent.

In smart home and personal applications, Myrizakis and Petrakis (2019) demonstrated fog-cloud integration using rule-based logic and RESTful services for home automation. Malik et

al. (2025) presented a cloud-controlled smart gardening system using the Blynk dashboard for monitoring and logging real-time environmental data. While these systems focused more on user interaction and condition-based triggers, they showed how cloud-IoT integration could be scaled to domestic and consumer-facing scenarios.

Although performance metrics were reported in only a few studies, all implementations showed high to medium-high automation capabilities, relying on either machine learning or rule-based engines. These studies highlight the potential of cloud-IoT systems to deliver intelligent, event-responsive automation across different operational scales, with fog and hybrid architectures often used to balance latency, reliability, and scalability.

## 2.4 Operational Efficiency Through Real-Time Monitoring

Real-time monitoring technologies have shown measurable impact in improving operational efficiency across small-to-medium service sectors, particularly laundromats, logistics, and waste management. These systems integrate Internet of Things (IoT) devices, machine learning (ML), and cloud platforms to optimize scheduling, reduce idle time, and improve resource utilization through live feedback loops and predictive analytics.

In the laundromat sector, Portase et al. (2024) developed SmartLaundry, a real-time monitoring system using IoT sensors and ML models such as random forest and XGBoost to manage public laundry allocation. Their system demonstrated scheduling prediction accuracy between 79.1% and 91.3%, reducing waiting times and enabling dynamic load balancing. Similarly, Kusrini et al. (2024) presented an AI-integrated self-service laundry kiosk, applying gradient boosting and embedded sensors for queue estimation and real-time monitoring, though without quantified outcomes. Menachery and Johnson (2021) utilized smartphone magnetic field sensors and cloud integration to monitor machine status in shared laundries, focusing on low-cost deployment and system robustness.

Logistics systems also benefited from real-time IoT applications. Brochado et al. (2024) proposed a modular IoT architecture for multimodal transport scheduling, employing WiFi/GPS, APIs, and constraint programming solvers. Their simulations yielded a Technical Efficiency Index ranging from 1.0 to 3.0. Amelia and Hudaya (2024) implemented IoT and ML in a medium-scale logistics firm, reducing delivery time by 15%, cost by 8%, and downtime by 20%. Wang and Wang (2024) developed a tracking system using Yolov5 and DeepSort for vehicle monitoring, achieving 95% utilization and minimal prediction errors (RMSE 0.25; MAE 0.20).

Other sectors illustrated further gains. In municipal waste collection, Bakhshi and Ahmed (2018) used ultrasonic sensors and ML analytics to reduce fuel use by 46% and collection time by 18%. Trihinas et al. (2018) applied adaptive monitoring across transport and wearable contexts, achieving a 74% reduction in data transmission and 71% energy savings while maintaining over 89% accuracy. Their system also extended device battery life by up to four days.

Across studies, key enabling factors included AI-based prediction, ML scheduling models, cloud integration, and scalable IoT architectures. However, limitations such as lack of large-scale deployments and reliance on simulation data reduce the generalizability of findings. Still, the results underscore the role of real-time monitoring in reducing operational waste, increasing process transparency, and supporting intelligent resource allocation in service-based systems.

## 2.5 Foundations of Smart Service Systems: The Role of Cloud, Wireless Sensor Networks, and Data Analytics

Cloud computing, wireless sensor networks (WSNs), and data analytics form the technological foundation for modern smart service systems. These enabling technologies support real-time data exchange, large-scale deployment, and intelligent decision-making across domains such as smart cities, healthcare, transport, and industrial monitoring.

There is a system that demonstrated how integrating WSNs with cloud platforms and Software Defined Networking (SDN) enabled self-managing environmental monitoring systems. By applying reinforcement learning, the system achieved an 85% reduction in transmission volume, thereby improving energy efficiency. Similarly, a 2018 editorial on smart sensory platforms highlighted mobile cloud integration and particle swarm optimization (PSO), resulting in 95.3% event classification accuracy and 93.3% presence detection in smart city environments.

In healthcare, Oğur et al. (2022) developed a real-time analytics system using Kafka, Spark, MongoDB, and RFID-enabled WSNs. The system attained 95% accuracy in health status processing, indicating the reliability of big data frameworks in critical environments. For urban transport, Khan et al. (2020) proposed a modular middleware system capable of handling over 25,000 clients per minute with sub-395 millisecond latency and 99.21% classification accuracy, highlighting the role of distributed caching and machine learning.

Several studies focused on system design and simulation to assess scalability and responsiveness. Pérez and Salvachúa (2021) 15odelled city-scale deployments using containerized WSANs, while Chenaru et al. (2015) explored RESTful services and Ipv6 for secure industrial WSN-cloud integration. Pandithurai et al. (2023) emphasized secure real-time environmental monitoring via IoT protocols such as GPRS, TLS, and Oauth2, though without performance metrics. Elkhoukhi et al. (2025) proposed a data stream processing architecture using Apache Kafka and SAMOA for low-latency smart building control, integrating ML into edge and cloud workflows.

Despite differences in domain, most architectures followed hybrid integration patterns, often combining edge/cloud intelligence with WSNs or IoT frameworks. Studies also identified operational challenges, including latency, interoperability, data volume, and integration complexity. Mishra et al. (2020), for instance, documented the latency-security trade-offs in smart home environments and stressed the importance of configurable cloud service models.

Overall, the literature affirms that these enabling technologies when integrated through modular, scalable, and secure architectures can deliver responsive and intelligent smart service systems. However, real-world scalability remains limited by a lack of empirical deployments and standardized performance reporting.

## 2.6 Challenges in Implementing Smart Laundromat Management Systems

The implementation of smart laundromat systems, particularly within small-to-medium enterprises (SMEs), is often hindered by technical, operational, and adoption-related challenges. While IoT-based automation promises increased efficiency and convenience, case studies and exploratory research reveal that practical deployments face significant barriers to scalability, user trust, and profitability.

Wolf et al. (2020) examined a failed implementation of a pay-per-use smart laundry machine, which encountered difficulties in achieving profitability. The study highlighted that despite the technical feasibility of the system, the business model was not sustainable, ultimately leading to implementation failure. The authors noted the importance of aligning smart service design with viable economic structures. However, the study did not explicitly discuss technical limitations such as hardware, connectivity, or system integration, limiting insights into the infrastructural barriers.

Roe et al. (2022) conducted an exploratory study on IoT deployment in smart retail settings, identifying device heterogeneity, security vulnerabilities, and service design flaws as major challenges. The complexity of managing a diverse range of devices led to scalability limitations, while privacy and security issues hindered user trust and compliance. These findings have direct implications for laundromat systems, where multiple IoT components must interoperate securely and reliably. Furthermore, Roe et al. emphasized that insufficiently designed data sharing environments reduce customer confidence and limit adoption, underlining the need for privacy-respecting architectures.

Both studies emphasized that the success of smart service systems depends not only on technical functionality but also on user-centered service design and sound business strategies. Service quality emerged as a cross-cutting issue Wolf et al. (2020) implicitly pointed to service expectations as a cause for failure, while Roe et al. (2022) explicitly linked privacy protection and security to service trustworthiness. Notably, user acceptance, maintenance issues, and cost of ownership were not detailed, suggesting a gap in the available literature regarding these practical implementation factors.

From a strategic standpoint, the reviewed studies suggest that smart laundromat initiatives require more than IoT device deployment. They must address interoperability, system scalability, and customer-facing concerns such as transparency, trust, and value delivery. Without attention to these dimensions, even technically sound systems may fail to deliver sustainable or scalable outcomes.

## 2.7 Comparative Frameworks in Industrial IoT

The integration of Industrial Internet of Things (IIoT) into control systems has significantly reshaped industrial automation. Boyes et al. (2018) proposed an IIoT analysis framework that classifies IIoT components based on device location, connectivity, user type, and criticality. The framework responds to the limitations in existing IoT taxonomies by providing a multi-dimensional structure suitable for analysing threats, vulnerabilities, and deployment contexts. This framework is particularly relevant for systems such as smart laundromats, where multiple device types and user interfaces interact across cloud and edge environments.

In comparison, Abuarqoub et al. (2017) introduced an integrated smart city framework that emphasized the fusion of IoT data using semantic and context-aware technologies. Their study focused on resolving interoperability and real-time responsiveness issues through a middleware-based architecture. Unlike Boyes et al., who emphasized system architecture and risk analysis, Abuarqoub et al. addressed data heterogeneity and latency in urban IoT environments. While both works agree on the need for distributed, modular designs, the smart city approach favours layered integration through semantic models and cloud-driven decision making, which could be adapted to industrial systems facing similar scalability and coordination challenges.

These two perspectives offer a strong foundation for comparative analysis. Boyes et al.'s framework is hardware-centric and fits systems with tight control and safety requirements, while Abuarqoub et al.'s model favours adaptive and intelligent environments using flexible software agents. The proposed smart laundromat management system may benefit from combining the security and categorization of IIoT assets described by Boyes et al. with the data integration and decision-making responsiveness emphasized by Abuarqoub et al. As laundromat machines become smart assets, managing interoperability, edge-cloud coordination, and control latency are critical. The comparative lens shows that selecting appropriate IIoT design patterns depends on balancing device-level control with high-level semantic integration and real-time feedback.

# CHAPTER 3

## METHODOLOGY AND PROJECT WORK

### 3.1 Project Development Framework

This project adopts the Agile Scrum approach as its development framework. Scrum is especially appropriate for software projects characterised by changing user requirements, hardware-software interactions, and the necessity for frequent testing and feedback. The hybrid Smart Laundromat Management System, which combines cloud platforms, mobile interfaces, and IoT devices, benefits from Scrum's structured yet adaptable methodology for managing iterative development cycles, reducing technical risk, and ensuring alignment with stakeholder requirements.

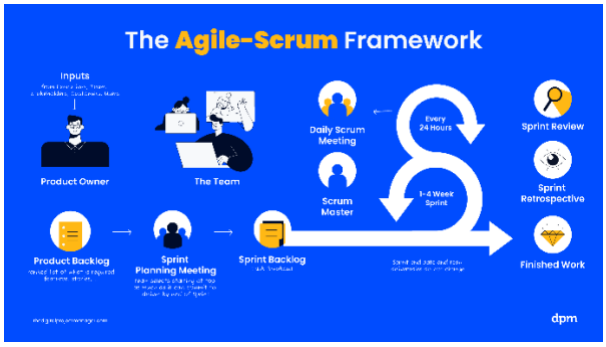



*Figure 3.1 Agile – Scrum Framework*

Each development sprint was scheduled for a four-week cycle, during which core modules were conceived, implemented, tested, and refined according to functionality objectives. Consistent sprint reviews allowed the team to evaluate progress, update task backlogs, and resolve technical or integration issues. The Scrum framework was essential for handling interdependencies among the mobile application, Firebase backend, and IoT hardware, necessitating regular synchronisation and incremental validation. The methodology consists of five major phases: Requirement Gathering and Analysis, System Design, Development, Testing, and Deployment.

### 3.1.1 Requirement Gathering and Analysis

The preliminary stage entailed identifying practical operational difficulties in traditional laundromat services. A synthesis of informal interviews, direct observations, and internet user evaluations was employed to pinpoint functional deficiencies and user grievances. Common problems included insufficient awareness regarding machine availability, the incapacity to pre-book machines, and ineffective queue management, particularly during peak periods. Business proprietors articulated a necessity for enhanced monitoring of machinery utilisation and revenue performance. Two principal user roles were established based on these findings:

- Customer: an individual utilising a laundromat who aims to verify machine availability, schedule reservations and obtain alerts.
- Owner: an administrator or business operator accountable for overseeing equipment utilisation, managing revenue records, and analysing system metrics.

Functional and non-functional needs were documented for both positions. Essential requirements encompassed real-time monitoring of machine status, secure login and registration, control for locking and unlocking machines, booking logic with temporal validation, and data synchronisation between mobile and backend systems. Technical specifications were defined for hardware components such as Arduino board, relays, and Bluetooth modules while for software frameworks such as Firebase Authentication, Firestore, Cloud Functions, and Flutter mobile framework.

### 3.1.2 System Design

The system design complies to a cloud-based client-server concept. The mobile application functions as the front-end interface for interactions between customers and owners. It interfaces with the backend housed on Google Firebase, which oversees user data, machine availability, booking records, and backend operations. Furthermore, Bluetooth communication facilitated localised machine control between the mobile application and the IoT devices.The system was segmented into logical units utilising UML diagrams, comprising:

- Use Case Diagrams: to outline the functions available to each actor which are customer and owner.
- Sequence Diagrams: to show interaction flow among the app, Firebase backend, and IoT devices during tasks such as booking, session completion, and unlocking.
- Component Diagram: to illustrates the modular design of the system, highlighting key elements such as the Bluetooth module, Firebase services, mobile app, and IoT device, as well as how they work together to provide system functionality.

Control and data flow diagrams were employed to illustrate the links among functional units. The system features role-based access management, allowing customers to view and reserve available machines, while owners possess enhanced privileges, including comprehensive analytics and machine usage information.

The machine locking control logic was designed to transmit a signal from the mobile application via Bluetooth to the Arduino microcontroller upon the completion of a session. This signal activates a relay linked to the simulated machine's power line, either securing or releasing it.

### 3.1.3 System Development

This phase involved parallel development efforts across the mobile application, backend services, and IoT hardware.

#### 3.1.3.1 Mobile Application Development

The mobile app was developed using the Flutter framework, allowing for rapid cross-platform development and UI flexibility. Key interfaces developed include:

- User registration and login screens: integrated with Firebase Authentication for secure access.
- Home dashboard: displays real-time machine availability.
- Booking interface: allows users to select time slots and machines.
- Admin dashboard: accessible only to verified laundromat owners, showing machine usage frequency, revenue analytics, and user trends.

Data storage and synchronisation were accomplished with Firestore, with documents representing users, machines, and bookings. Reservations are time-stamped and verified against the current operational status of machines prior to confirmation, so averting scheduling conflicts.

#### 3.1.3.2 Cloud Backend Development

The Firebase platform was utilized for backend development due to its scalability, real-time data handling, and compatibility with Flutter. Key components included:

- Authentication: Firebase Authentication handled user login and role-based access.
- Firestore Database: Used to store machine metadata, booking records, and user activity logs.
- Cloud Functions: Implemented to automate backend tasks such as:
- Updating machine status after a session timeout.
- Sending push notifications when a machine becomes available.
- Resetting lock states to default in the event of a system error.

#### 3.1.3.3 IoT Hardware Integration

The hardware implementation utilized an Arduino Uno board, equipped with:

- A Bluetooth HC-05 module to receive serial commands from the mobile app.
- Relay modules simulating power control (lock/unlock) of washing machines.
- A simple LED indicator to represent machine status (active/idle).

Communication latency was optimized to achieve round-trip feedback within 2–3 seconds. The Arduino also included a basic acknowledgment protocol to confirm successful execution of commands, improving reliability.

### 3.1.2 Testing

To ensure system effectiveness, multiple testing levels were performed:

#### 3.1.2.1 Prototype Testing

Each component was tested independently. The mobile app's UI was validated for responsiveness and correctness. Firebase Authentication and Firestore operations were tested using mock users. The Arduino board was tested with manual commands to verify relay switching and Bluetooth stability.

#### 3.1.2.2 User Flow Testing

Simulated booking scenarios were tested to validate session conflict handling, such as overlapping reservations or expired bookings. The transition between booked and available states was monitored for accuracy. The admin dashboard's ability to reflect usage data in near real-time was also verified.

#### 3.1.2.3 System Integration Testing

The complete system application, backend, and hardware was tested in an end-to-end configuration. Booking a machine triggered a state change in Firestore, which prompted the app to send a command to the Arduino, activating the relay. Successful feedback confirmed command reception, and all actions were logged in Firestore.

Edge cases were also tested, such as:

- Network disconnection during a booking.
- Unexpected power loss on the Arduino.
- Duplicate session attempts.

### 3.1.3 Deployment

Following successful testing, the system was deployed in a test environment. The mobile application was installed on Android test devices and connected to a live Firebase backend. The hardware prototype was assembled using mock washing machines represented by switchable relay-controlled devices.

The final prototype simulated a mini laundromat setup where users could interact with the app to:

- Register and log in.
- View machine availability.
- Book a machine and receive confirmation.
- Control the lock/unlock status of the machine.

The deployment demonstrated that the system could handle real-time interactions between software and hardware components with minimal delay and high reliability.The implementation of the Scrum approach facilitated efficient time-boxing, modular development, and incremental feature delivery. Consistent feedback mechanisms throughout sprint reviews guaranteed that execution stayed in accordance with technical objectives and user anticipations. The project developed a functional prototype that resolved significant operational issues in the laundry sector by incorporating IoT, cloud computing, and mobile technologies into a systematic agile process.

## 3.2 Tools And Technologies Use

This project uses multiple software and hardware components to build a functional and responsive smart laundromat management system. Each tool was selected based on its technical compatibility, ease of integration, cost-effectiveness, and support for real-time communication. The following section outlines and explains the tools and technologies used across mobile development, cloud services, and embedded hardware.

### 3.2.1 Software Tools

*Table 1 Project Software Tools*

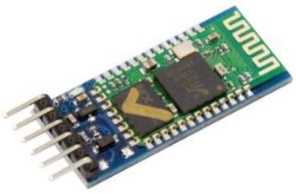
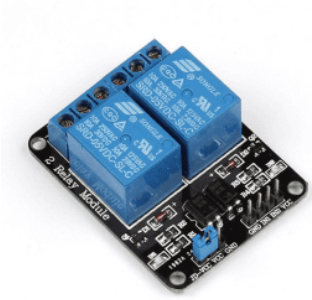
| Tools | Purpose of Usage |
|---|---|
| Flutter | Flutter is an open-source user interface (UI) development framework maintained by Google. It allows developers to build high-performance, cross-platform applications using a single Dart codebase. In this project, Flutter is used to develop the mobile application interface for both customers and laundromat owners. The framework enables rapid |

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| | development and real-time preview of UI components. Its widget-based architecture simplifies the creation of interactive pages such as login, machine availability, booking status, and administrative dashboards. Flutter's built-in support for Bluetooth plugins and Firebase integration further justifies its selection for this project. |
|  | Firebase Authentication is a secure and scalable user identity platform that supports various authentication methods, including email-password, phone number, and third-party OAuth providers. It ensures secure login and access control in the mobile application. In this system, Firebase Authentication is used to distinguish between two types of users: customers and laundromat owners. This role-based authentication ensures that sensitive features, such as viewing machine usage statistics and revenue data, are restricted to verified business owners. |
|  | Cloud Firestore is a NoSQL, cloud-hosted database offered by Google Firebase. It allows for real-time data synchronization and querying, making it suitable for systems requiring low-latency performance and dynamic data updates. Firestore stores all critical data structures in this project, including user profiles, machine statuses, booking records, and system logs. Its document-based architecture enables flexible schema design, which is essential when managing varying data attributes across different system modules. Firebase Cloud Functions provide a serverless backend environment where custom logic can be executed in response to specific events within the Firebase ecosystem. In the smart laundromat system, Cloud Functions are used to automate backend tasks such as updating machine availability after a booking session ends, sending push notifications when a machine becomes free, and resetting machine lock states during system |

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| | restarts. This reduces the need for manual intervention and ensures that critical tasks run consistently in the background. |
|  | The Arduino Integrated Development Environment (IDE) is an open-source platform used to write and upload code to Arduino microcontroller boards. It supports the C and C++ programming languages and offers built-in libraries for handling Bluetooth, serial communication, and GPIO operations. In this project, the Arduino IDE is used to program the Arduino Uno board to interpret lock/unlock commands sent from the mobile application via Bluetooth, activate relays accordingly, and return feedback to the app. |

### 3.2.2 Hardware Components

*Table 2 Project Hardware Components*

| Tools | Purpose of Usage |
|---|---|
|  | The Arduino Uno is an open-source microcontroller board based on the ATmega328P chip. It offers 14 digital I/O pins and 6 analog inputs, making it suitable for basic control tasks and hardware interfacing. In this project, the Arduino serves as the central controller that receives lock/unlock commands from the mobile application via Bluetooth and triggers the relay modules to simulate machine control. Its simplicity, affordability, and wide developer community make it a practical choice for academic projects involving IoT. |
|  | The HC-05 is a Bluetooth serial module designed for short-range wireless communication. It enables two-way serial data transfer between the mobile device and the Arduino board. In this system, the HC-05 facilitates real-time communication by receiving control signals and transmitting confirmation back to the app. It operates on the Bluetooth 2.0 protocol and is configured using AT commands to set the desired mode and baud rate. Its low power consumption and ease of integration with the Arduino make it ideal for real-time, device-level interaction. |
|  | Relay modules are electromechanical switches that allow low-power digital signals to control high-voltage electrical circuits. In the smart laundromat prototype, relay modules are used to simulate the power control mechanism of washing machines. When the user completes a booking, the relay is activated via the Arduino to lock the machine. Upon session completion or manual release, the relay is toggled to unlock the machine. This design reflects how an actual |

| | |
|---|---|
| | laundromat could manage machine access in real deployments. |
|  | LEDs are used as visual status indicators in the hardware prototype. A red LED may indicate that a machine is currently booked or locked, while a green LED signals availability. These indicators assist in debugging and enhance user feedback during system demonstration and testing. |
|  | A regulated DC power supply is used to power the Arduino and relay circuits. A solderless breadboard allows rapid prototyping and circuit testing without permanent soldering. Jumper wires and resistors are used to establish stable and flexible circuit connections. This setup supports safe and repeatable testing during development. |
|  | An Android device is required to install and test the Flutter-built mobile application. This device is used to verify user login, booking functionality, Bluetooth pairing, real-time system control, and admin view rendering. Android was chosen over iOS due to Flutter's broader plugin compatibility and simpler Bluetooth configuration on Android platforms during development. |

## 3.3 Sprint 1

The initial sprint constructed the project's foundation, concentrating on gathering requirements and analysing the problem context within traditional laundry settings. This phase was essential in recognising both technical and user-focused requirements, which would inform all future design and development choices. The team employed a systematic method to collect feedback from genuine laundromat consumers and company proprietors. Informal interviews and observational techniques were utilised to identify critical issues, including erratic machine availability, queue congestion, the absence of reservation choices, and insufficient operational insights for laundromat proprietors.

The user role has been identified and specified which are customer and owner, to formalise the issue domain. The customer role included persons utilising the laundromat machines, necessitating fundamental characteristics such as real-time availability assessments, reservation capabilities, and session alerts. The owner role pertains to the system administrator or business operator requiring access to analytics, usage logs, and machine performance statistics. This differentiation influenced initial choices regarding authentication, access control, and interface design.

Subsequent to stakeholder analysis, a detailed listing of functional and non-functional requirements was compiled. Functional requirements included safe user registration and authentication, real-time machine surveillance, booking administration, and machine locking/unlocking control. Non-functional requirements, including system stability, data consistency, real-time responsiveness, and user-friendliness, were also recorded. A rough product backlog was concurrently established, enumerating all anticipated features and components in a prioritised format. This backlog functioned as a dynamic document to direct development in subsequent sprints.

Sprint 1 required the selection of the technology stack for both the software and hardware components of the project. Flutter has been chosen for mobile development, Firebase for cloud backend services, and Firestore for real-time database management in the software stack. The Arduino Uno was selected as the microcontroller, accompanied by a relay module and Bluetooth HC-05 for machine control and wireless communication. The interoperability,

cost-effectiveness, and supportive community associated with these technologies rendered them exceptionally appropriate for academic and prototype applications.

In this sprint, the development commenced the configuration of the Firebase project and initiated the setup of the Flutter development environment. This preliminary effort facilitated a more seamless transition into initial prototyping and design during Sprint 2. Sprint 1 established the essential understanding and planning required to synchronise technical execution with user expectations and corporate goals, guaranteeing that subsequent development would be both purposeful and technically viable.

## 3.4 Sprint 2

Sprint 2 concentrated on converting the requirements collected in the prior phase into a formalised system architecture and initial functional prototypes. The main goal of this sprint was to develop a comprehensive technical design for the Smart Laundromat Management System and to initiate the validation of hardware-software integration through physical prototypes. This sprint signified the shift from planning to concrete implementation, establishing the foundation for comprehensive development in subsequent sprints.

The initial significant task was the creation of Unified Modelling Language (UML) diagrams to delineate system behaviour and component interaction. Use case diagrams were developed to delineate the functional scope for each user job. For clients, these applications encompassed seeing available machines, reserving a session, and receiving notifications. Owners utilised the application for monitoring metrics and maintaining booking histories. Class diagrams were created to delineate the structure of essential entities, including Users, Machines, and Bookings. Sequence diagrams were created to represent time-based interactions across the mobile application, the Firebase backend, and the IoT hardware, particularly for events like starting a booking, activating a machine, and releasing it upon completion.

The system architecture design was completed concurrently. The solution utilises a cloud-based client-server architecture in which the mobile application interacts with Firebase services for authentication, data storage, and backend processing. Furthermore, Bluetooth connection was chosen to link the mobile application with the Arduino-based IoT hardware. This architecture facilitates modular development and enables the application to operate in both online and local environments.

The Firestore database structure was drafted and largely implemented in this sprint. Collections were created to store user profiles, machine data, booking records, and real-time system logs. Each document was organised with fields including machine ID, booking time, user ID, and status indications. This design guaranteed rapid real-time data querying, essential for functionalities like live availability verification and conflict mitigation.

Simultaneously, the initial prototyping of the hardware components commenced. An Arduino Uno microcontroller was programmed to receive serial commands through an HC-05 Bluetooth module. A fundamental relay module was linked to emulate the locking mechanism of a washing machine. The breadboard circuit was constructed to evaluate if commands transmitted via a terminal emulator or test script could consistently activate the relay to flip ON or OFF. LED indicators were used to visibly display the lock status.

The Flutter development environment was established to incorporate packages necessary for Bluetooth communication in order to commence the integration of the app with the hardware. Preliminary interface wireframes were developed for screens like machine availability and booking confirmation. Although the complete user interface was not yet operational, the foundation was established for linking Flutter UI actions to hardware commands.

By the conclusion of Sprint 2, the project had developed a validated architectural framework, a comprehensive grasp of data linkages, and a functional hardware prototype capable of simulating real-world processes. This sprint was crucial in validating the system's capability for bidirectional interaction between the software and physical components, so allowing the project to advance safely into comprehensive application development in the subsequent sprint.

## 3.5 Sprint 3

Sprint 3 focused on the creation of the essential capabilities of the mobile application, serving as the major interface between consumers and the smart laundry system. This sprint signified the shift from design and low-level hardware prototypes to user-centric software development. The mobile application was created utilising the Flutter platform, facilitating swift interface creation and effortless interaction with the Firebase backend. This sprint's primary deliverables encompassed user registration and login processes, machine discovery functionalities, and booking activities, all supported by real-time database synchronisation.

The initial milestone of this sprint was the execution of user authentication with Firebase Authentication. The login and registration modules were developed to enable users to register using email and password credentials. Role differentiation was mandated at registration by necessitating users to choose either “customer” or “owner.” After a successful login, users were routed to dashboards tailored to their roles. Customers used a machine booking interface, whilst owners were granted entry to a secured area designated for future development into an analytics dashboard.

Subsequently, the machine availability interface was developed. This interface retrieved real-time data from Firestore to exhibit the current state of all machines. The data was presented in a card-based format, using colour-coded indicators and iconography to enhance user understanding. Real-time listeners were established utilising Firebase’s onSnapshot methods, allowing the application to instantly display state changes anytime a booking was created, modified, or cancelled by any user.

The booking system was subsequently introduced. Customers were permitted to choose a machine, specify a session duration, and finalise the reservation. Upon confirmation, the booking information was recorded in Firestore, which includes fields such as machine ID, user ID, start time, end time, and status. The application used logic to avert booking conflicts by confirming machine availability before to submitting a new reservation. The booking buttons were automatically disabled for machines that were either in use or reserved for conflicting time slots. These real-time verifications were crucial to maintain data integrity and prevent overbooking.

Alongside fundamental booking capabilities, session monitoring and user feedback were partially included. A booking confirmation interface was established to present session particulars, encompassing expected completion time and lock status. The foundation was established for subsequent connection with the IoT module, enabling the application to later activate the machine lock using Bluetooth.

In this sprint, the Firebase Firestore schema was revised and evaluated to provide seamless data operations. Firestore security rules were established to limit unauthorised access to booking and machine documents. For instance, customers were restricted to reading machine statuses and modifying their own booking records, whereas owners were afforded more access for reporting and monitoring functions.

Sprint 3 contained preliminary usability testing of the interface. Test accounts were established to replicate reservations from various users. Real-time updates confirmed that system statuses changed without the need for manual refresh. The findings indicated that the mobile application effectively functioned as a real-time control interface, precisely handling user accounts and reservations.

At the conclusion of Sprint 3, the project successfully developed a fully operational booking interface utilising a real-time cloud backend. This sprint developed the application's functional core and validated the technical feasibility of employing Firebase as a real-time database solution for booking administration. These capabilities established the groundwork for sophisticated logic, analytics, and hardware integration in subsequent iterations.

### 3.6 Sprint 4

Sprint 4 focused on backend automation and the creation of the administrative interface. Following the completion of essential user-side functionalities in the preceding sprint, focus shifted to enhancing system intelligence via automated workflows and delivering actionable insights to laundromat owners. This sprint focused on integrating Firebase Cloud Functions and implementing a dedicated admin dashboard. These components were crucial for overseeing machine sessions autonomously and guaranteeing the system's capacity for sustained operational efficiency.

The initial component discussed was backend automation via Firebase Cloud Functions. Cloud Functions were set up to do event-driven tasks upon the fulfilment of particular conditions in Firestore. A significant automation was the automatic machine release, activated at the conclusion of a booking period. This function tracked each booking document and revised the associated machine state from "booked" to "available" upon session expiration. This eliminated the necessity for user-initiated session termination and guaranteed uniform updates on machine availability.

A separate Cloud Function was created to deliver real-time push alerts to users via Firebase Cloud Messaging (FCM). These messages alerted users when a reserved machine became available or when a booking session was nearing its conclusion. These warnings increased user engagement and minimised idle time between machine usage sessions. A fail-safe protocol was established to identify any anomalous session lengths (e.g., exceeding anticipated time) and notify the owner for human review.

A separate owner dashboard was integrated into the mobile application alongside backend automation. This interface was limited to authorised users with the "owner" role, as confirmed by role-based claims established at registration. The dashboard exhibited essential performance metrics, including the total finished sessions, frequently utilised machines, average booking lengths, and projected revenue. The metrics were obtained from Firestore using server-side aggregation and presented using chart widgets and tabular summaries in the Flutter application.

The Firestore data schema was augmented to incorporate usage history logs to facilitate these functions. Each finalised booking was added to a specific history collection, facilitating longitudinal data analysis. This facilitated the system's ability to produce temporal trends and provided owners with a comprehensive assessment of business performance. The security protocols were revised to let only people with owner-level access to obtain sensitive financial and operational information.

Throughout this sprint, the application underwent systematic testing to verify the accuracy of automated features. Test reservations were generated with simulated start and finish times to ascertain the accurate designation of machine availability and the proper transmission of notifications. Owner accounts were evaluated for unauthorised access attempts to verify that Firebase policies inhibited privilege escalation.

By the conclusion of Sprint 4, the system had progressed from fundamental functionality to a more advanced, event-aware service. The backend logic could autonomously monitor time-sensitive procedures, allowing laundromat owners to obtain a dependable, real-time summary of machine utilisation and revenue production. These developments improved the system's feasibility for real-world implementation and facilitated the integration of hardware control elements in the subsequent sprint.

## 3.7 Sprint 5

Sprint 5 indicated the integration of Internet of Things (IoT) hardware with the smartphone application, facilitating real-time physical management of laundry machines. This phase concentrated on integrating the digital and physical components of the system, after the establishment of the software infrastructure and user interface in previous sprints. The main goal was to establish uninterrupted connection between the mobile application and Arduino-based controllers using Bluetooth, enabling authorised users to lock or unlock devices in real time according to their booking status.

The integration commenced with the configuration of the Arduino Uno microcontroller, which functioned as the physical controller for each washing machine unit. The Arduino was configured with the Arduino IDE to receive serial input via the HC-05 Bluetooth module. Upon receiving commands such as "LOCK" or "UNLOCK," the Arduino would activate a relay module, simulating the power switching of the machine. An LED indicator was added in parallel to visually verify the lock state where red signifying locked and green signifying available.

The Flutter project was enhanced with Bluetooth communication modules, utilising third-party packages for device detection, pairing, and message delivery. The application was designed to automatically connect with the appropriate HC-05 module during a booking session and transmit control commands upon confirmation of the booking. The application was designed to exhibit machine status according to feedback from the Arduino, establishing a closed-loop system that enabled the user to confirm the successful execution of commands.

A latency benchmark was built during this sprint to ensure real-time reliability. The objective was to accomplish complete round-trip communication from application command to relay activation and back to application confirmation within three seconds. This benchmark was achieved in many test settings, with the majority of exchanges concluding in under two seconds. Command-response protocols were refined by the utilisation of succinct acknowledgement messages from the Arduino to guarantee that the application obtained confirmation prior to updating Firestore or notifying the user.

Error handling procedures were created to mitigate potential Bluetooth communication failures. This included retry mechanisms for unsuccessful transmissions, user alerts for pairing inconsistencies, and emergency protocols recommending users to reach out to the owner in cases of recurrent lock failures. These enhancements improved the system's resilience, particularly in scenarios when signal interference or device mispairing could compromise reliability.

Simultaneously, security and control borders were strengthened. The lock/unlock functionality was limited to users possessing active and valid reservations. This was implemented via both the mobile application interface and the command transmission protocol, where invalid or expired bookings would inhibit any command from being transmitted. Owners were granted override privileges via the admin dashboard to manually unlock the machine in the event of system failure or emergency.

By the conclusion of Sprint 5, the project attained comprehensive end-to-end management of physical machines via a mobile application, with real-time responsiveness and status updates. The successful conclusion of this sprint confirmed the viability of incorporating an economical IoT solution into a cloud-based laundry management system. With hardware control functioning reliably and securely, the system was prepared for comprehensive integration testing and deployment in the final phase.

## 3.8 Sprint 6

Sprint 6 represented the conclusion of all development efforts, integrating the mobile application, cloud backend, and IoT devices into a cohesive functioning prototype. The main objective of this sprint was to execute comprehensive system integration, verify functioning in real-world scenarios, rectify any discrepancies, and finalise the system for deployment and demonstration. In accordance with Scrum standards, this sprint included an assessment of the completed product against the initial requirements and sprint objectives, confirming that each user story had been addressed and delivered.

The integration procedure commenced with the establishment of synchronised communication among all system components. The mobile application interfaced with the Firebase backend for user authentication, machine status acquisition, and booking administration. Simultaneously, Bluetooth connectivity was evaluated to confirm that orders from the mobile interface were consistently received by the Arduino-based IoT hardware. Experiments were performed utilising authentic booking situations, wherein users reserved computers, initiated lock commands, finalised sessions, and monitored the system's automatic unlocking and status resets. These simulations confirmed the bidirectional data flow across all three layers: mobile, cloud, and hardware.

A crucial aspect of this sprint involved executing thorough system testing. This included unit testing of discrete modules, including authentication, booking form validation, and database queries. Integration testing concentrated on validating the whole operational flow, particularly the accuracy of booking lifecycle events, synchronisation of machine lock states, and the timing of Cloud Functions. The testing process included edge scenarios, such as concurrent booking attempts by various users, expired sessions, Bluetooth signal loss during locking requests, and sudden changes in system state. These scenarios evaluated the system's resilience and validated its capacity to uphold data integrity during simultaneous and asynchronous occurrences.

To validate system logging, Firestore event records were examined. The system documented every booking's creation, modification, and cancellation with precise timestamps. Correspondingly, machine status transitions were accurately represented in the database and

mobile interface. Notifications produced via Firebase Cloud Messaging were evaluated using simulated triggers and validated for their promptness and precision. Administrative users accessed booking records and revenue reports promptly, confirming the dashboard's real-time capabilities.

Following to successful testing, the team finalised the prototype deployment. The Flutter application was installed on an Android test device and set up to interface with the live Firebase project. A mini laundromat setup was constructed utilising simulated washing machines, LED indicators, relay modules and a compact Arduino circuit. This configuration enabled to exhibit system capabilities in a regulated yet authentic environment, encompassing live booking, physical machine control, and role-based access management.

Alongside technical validation, final adjustments and documentation were finalised during this sprint. The user interface was refined for clarity and responsiveness. The security protocols in Firebase were evaluated to avert unauthorised access to sensitive collections. A user manual was created to outline application usage, and test logs were recorded to substantiate the system evaluation section of the final report.

At the conclusion of Sprint 6, the smart laundromat management system was entirely operational, incorporating real-time cloud services, embedded hardware, and a mobile interface. The finalised prototype fulfilled all primary objectives, encompassing real-time monitoring, remote machine operation, user-specific reservations, and data logging. This sprint validated the technical viability of the solution and showcased its preparedness for practical implementation in small-scale laundry operations.

## 3.9 System Architecture

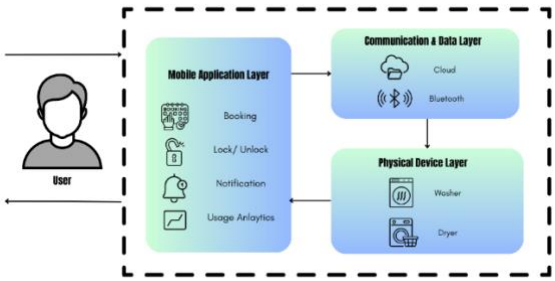



*Figure 3.9 System Architecture*

Figure 3.9 illustrates the system architecture of the Smart Laundromat Management System is organized into three primary layers: the Mobile Application Layer, the Communication and Data Layer, and the Physical Device Layer. These components interact to enable real-time monitoring, user interaction, and hardware control in a cloud-IoT integrated laundromat environment. The entire system is designed to enhance operational efficiency and user experience through automation, remote accessibility, and data-driven decision-making.

### 3.9.1 Mobile Application Layer

This layer serves as the user interface for both customers and laundromat owners. It is developed using Flutter and provides four main functions:

- Booking: Allows users to view available machines and reserve them in advance, thus reducing waiting time and preventing queue conflicts.
- Lock/Unlock: Users can remotely secure their machines after use to prevent unauthorized access, a feature available through Bluetooth integration.
- Notification: Push notifications are sent to users when washing or drying cycles are complete or if any activity occurs during machine use.
- Usage Analytics: Admin users can access visualized data on machine usage patterns, enabling informed decisions regarding operations and maintenance.

This layer bridges the interaction between the user and the underlying system by sending commands and receiving status updates from the hardware layer through the communication interface.

### 3.9.2 Communication and Data Layer

This layer handles all communication between the mobile app and the physical devices. It includes:

- Cloud Services: Firebase is used to store user data, machine availability status, booking records, and usage analytics. It ensures remote data accessibility and synchronization across devices.

- Bluetooth Module: A local communication channel between the mobile application and Arduino-powered IoT hardware. It is primarily used for sending lock/unlock commands and receiving status feedback in near real-time.

The combination of cloud storage and Bluetooth connectivity forms a hybrid communication model that supports both local and remote system interactions.

### 3.9.3 Physical Device Layer

This layer consists of the actual laundromat machines, such as washer and dryer. Each machine is connected to a microcontroller (e.g., Arduino) equipped with sensors and relay modules. These components receive operational commands (e.g., lock/unlock) from the Bluetooth module and send back status updates. This layer ensures the physical execution of user commands and provides real-world interaction with the system.

## 3.10 UML Diagrams

### 3.10.1 Use Case Diagram

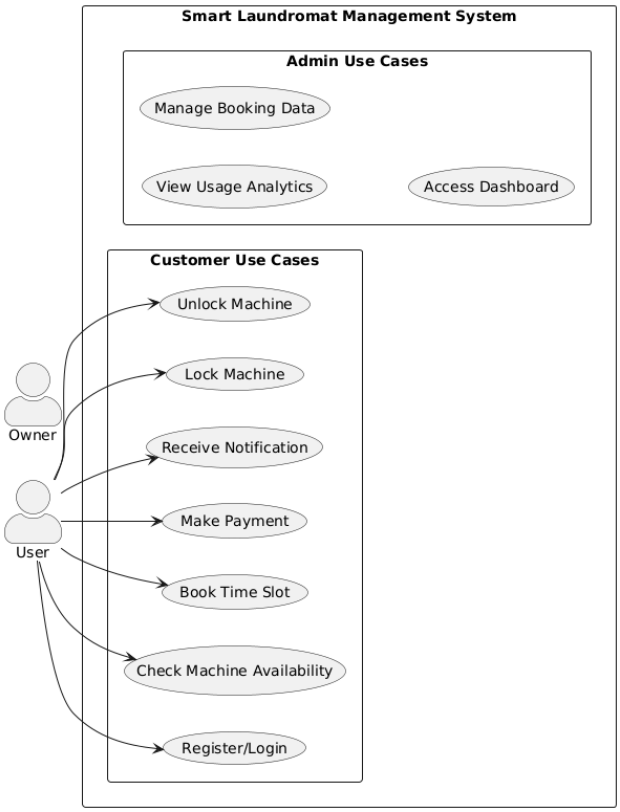



*Figure 3.10.1 Use Case Diagram*

Figure 3.10.1 shows the main interactions between two user roles, namely the User as the customer and the Owner as laundromat owner, with the Smart Laundromat Management System. The system is divided into two major functional areas: Customer Use Cases and Admin Use Cases. Customers can perform operations such as registering or logging into the system, checking machine availability, booking time slots, making payments, receiving notifications, and remotely locking or unlocking washing machines. On the other hand, the Owner has access to functionalities such as viewing usage analytics, managing booking data, and accessing the administrative dashboard. This diagram effectively captures the scope of functional requirements from both user types, aligning with the system's core objectives to improve service efficiency and operational visibility.

### 3.10.2 Sequence Diagram

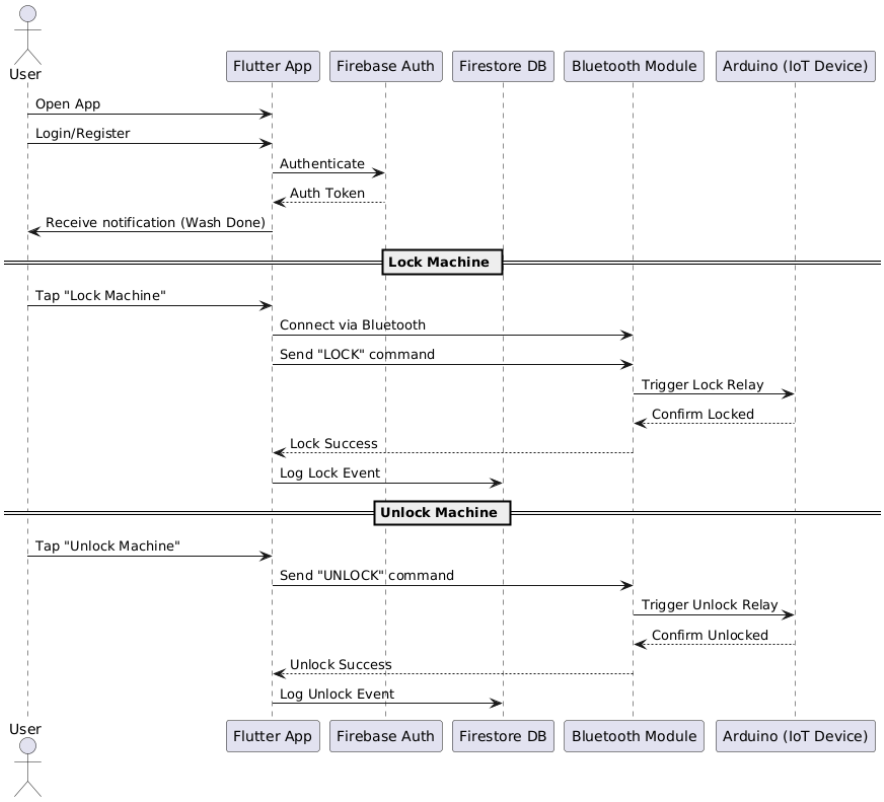



*Figure 3.10.2 Sequence Diagram*

As shown in Figure 3.10.2, it explained the procedural flow involved in the Booking, Locking, and Unlocking features of the system. The interaction begins when the user opens the application through log in and receives a notification upon wash completion. To secure their laundry, the user selects the “Lock Machine” option. This triggers a Bluetooth connection to the Arduino device, sending a lock command that is processed by the relay, followed by a confirmation. The event is logged in the system database. Similarly, the “Unlock Machine” command follows the same structure, sending an unlock instruction and receiving confirmation once the machine is accessible again. Each component in the diagram that include the mobile application, Firebase services, Firestore database, Bluetooth module, and Arduino device, plays a critical role in enabling secure, real-time machine control. The diagram confirms the alignment between functional requirements and system behaviour.

### 3.10.3 State Chart Diagram

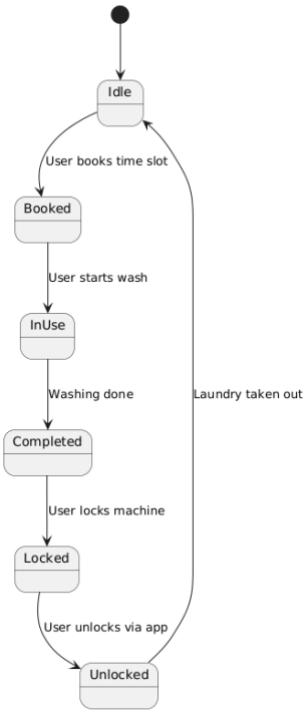



*Figure 3.10.3 State Chart Diagram*

Figure 3.10.3 illustrates the model's lifecycle of a laundromat machine within the system. It begins from the Idle state, indicating that the machine is not currently in use. When a user books a time slot, the machine transitions to the Booked state. Once the wash cycle begins, the machine enters the In Use state. After the washing process is completed, the machine moves to the Completed state. From there, the user may choose to lock the machine for added security, transitioning the machine to the Locked state. Upon arriving at the laundromat, the user can unlock the machine through the mobile application, which changes the state to Unlocked. Finally, when the laundry is collected, the machine returns to the Idle state, ready for the next cycle. This functional model helps clarify machine usage flow and system response to user actions.

### 3.10.4 Component Diagram

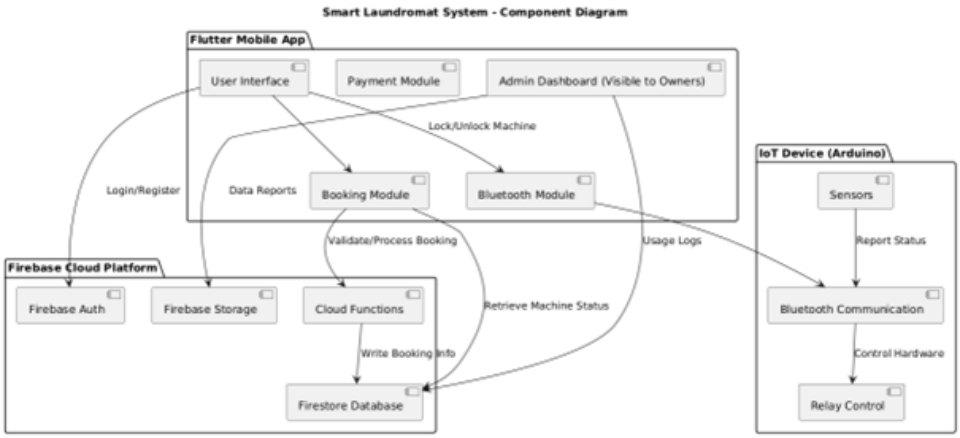



*Figure 3.10.4 Component Diagram*

Figure 3.10.4 illustrates the key modules of the Smart Laundromat System, which is divided into three main subsystems: the Flutter Mobile App, the Firebase Cloud Platform, and the IoT Device (Arduino). The mobile app acts as the user interface, allowing customers to log in, book machines, and control lock/unlock features through Bluetooth. It also includes an admin dashboard, accessible only to laundromat owners, to monitor usage analytics. The booking module manages reservations and communicates with Firebase to validate and store booking data.

Firebase Cloud Platform handles backend services such as authentication (Firebase Auth), data storage (Firestore Database), and automated processes (Cloud Functions). It ensures all user activities and machine statuses are logged and synced in real time. On the hardware side, the IoT Device consists of sensors to detect machine status and relays to control machine functions. Communication with the app occurs via a Bluetooth module, enabling command transmission and status feedback. This overall architecture supports real-time interaction between users, the mobile app, the cloud, and physical laundromat machines, forming an integrated smart system.

## 3.11 Proposed System Interface

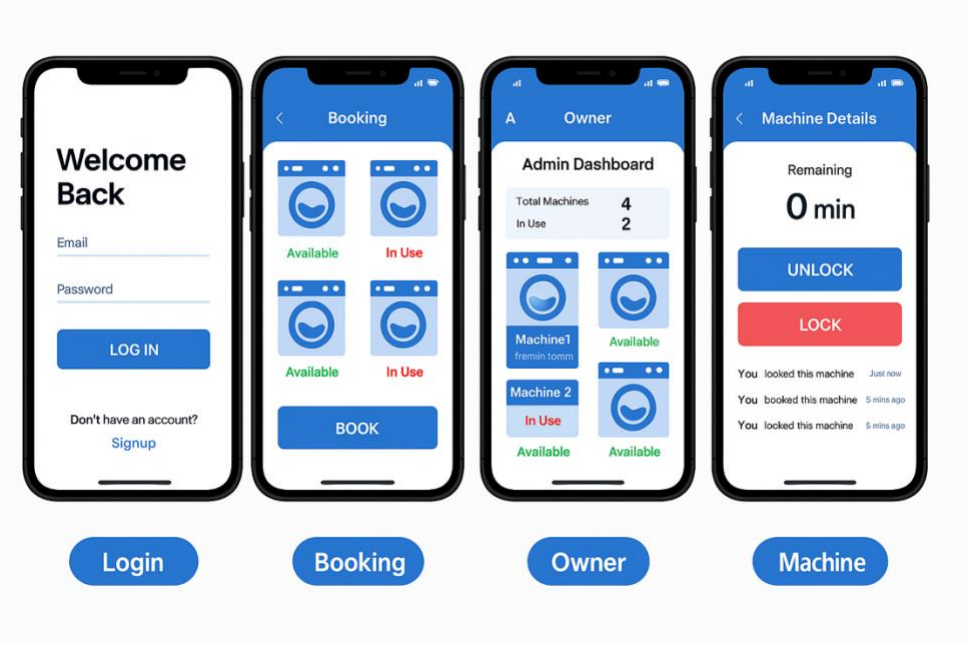



*Figure 3.11 System Interface*

Figure 3.11 shows the proposed design of four core screens of the mobile application that will be developed for the laundromat system. The first screen is the Login Page, which includes user authentication fields for email and password input. The second screen demonstrates the Booking Interface, where available and in-use machines are visually displayed, and users can make reservations by tapping the "Book" button. The third screen presents the Admin Dashboard, available only to owners, displaying machine status and control options. The final screen is the Machine Details Page, where users are given the ability to remotely lock or unlock a machine, and track booking and locking history. The user interface design emphasizes clarity, ease of navigation, and a clean layout suitable for mobile environments.

# CHAPTER 4

## RESULT AND DISCUSSION

### 4.1 Preliminary Survey

The preliminary survey was conducted to understand current issues in laundromat operations and to validate the problems outlined in the problem statement. The survey involved 30 respondents, consisting of both laundromat users and business owners. This structure provided a balanced view of customer experiences and operational challenges from the provider side. The findings informed the design of the Smart Laundromat Management System by identifying the core pain points that needed to be addressed through IoT and cloud integration.

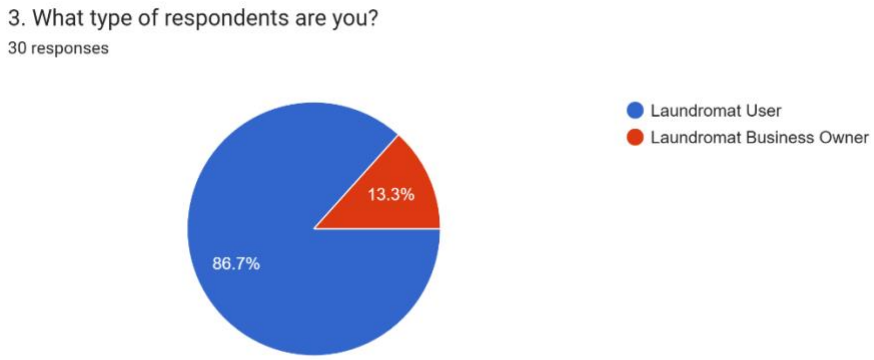



*Figure 4.1 Type of Respondents*

Figure 4.1 illustrates the distribution of respondent types. A large majority of respondents were laundromat users, representing 86.7 percent of the sample. Business owners made up the remaining 13.3 percent. This distribution reflects the real environment where customer volume is significantly higher, and it provides a strong foundation for evaluating consumer challenges. At the same time, input from owners ensures that operational concerns are captured and integrated into the system requirements.

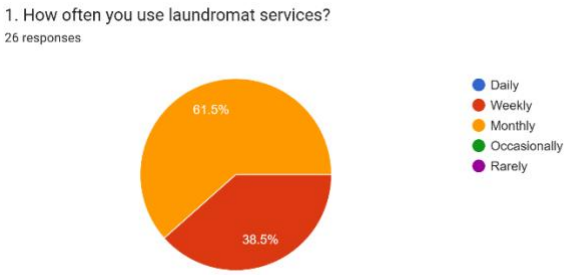



*Figure 4.2 Visit Frequency*

The survey also examined usage behaviour among users. Figure 4.2 shows that most users visit laundromats frequently, with 61.5 percent reporting monthly use and 38.5 percent reporting weekly use. There were no respondents who used laundromats daily or rarely in the dataset. This suggests that laundromat services remain a routine necessity for a large segment of the population, especially among those who lack in-house laundry facilities. The frequency of use highlights the need for a system that improves efficiency, reduces inconvenience and enhances transparency.

The survey explored the challenges commonly faced by laundromat users. Respondents reported uncertainty about machine availability as the biggest issue. Many users explained that they often arrive only to discover that all machines are occupied, which forces them to wait or return at a later time. Users also described frustration with the absence of estimated wait times and the inability to know when a machine would become free. These findings confirm that the lack of real time information significantly impacts user experience.

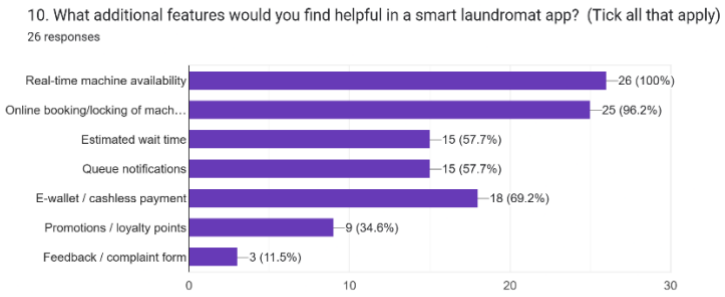



*Figure 4.3 Features in Smart Laundromat*

The survey then examined what users would find helpful in a smart laundromat system.

Figure 4.3 summarises the feature preferences of 26 user respondents. Real time machine availability was the most requested feature, selected by 100 percent of participants. This was followed closely by online booking, selected by 96.2 percent. Other desirable features included estimated wait time and queue notifications, each selected by 57.7 percent of respondents, and cashless payment, selected by 69.2 percent. These results show a strong demand for digital convenience and more organised machine usage. They also highlight that users desire transparency, reduced waiting time and better control over their laundry schedule.




*Figure 4.4 Numbers of Machines*

Business owners provided insight into operational conditions from the management perspective. Figure 4.4 shows the distribution of machine counts among the four owner respondents. Most laundromats in the dataset operated with more than twenty machines, with 75 percent falling in this category, while 25 percent operated between eleven and twenty machines. This variation indicates that laundromat size can influence how operational issues are experienced and managed.

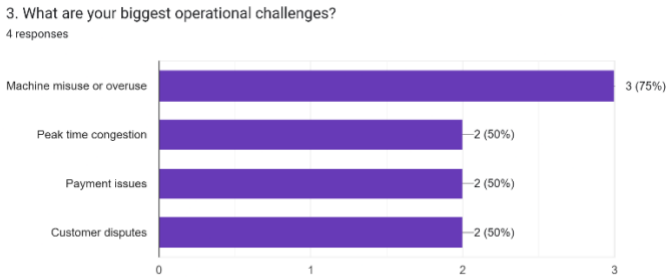



*Figure 4.5 Challenges for Owners*

Owners also identified several key operational challenges, as shown in Figure 4.5. Machine misuse or overuse was reported by 75 percent of the owners. Peak time congestion was mentioned by 50 percent, along with payment issues and customer disputes. These issues point to the lack of monitoring tools and structured systems for managing machine usage. They also reflect the need for analytics and automated tracking to improve oversight.

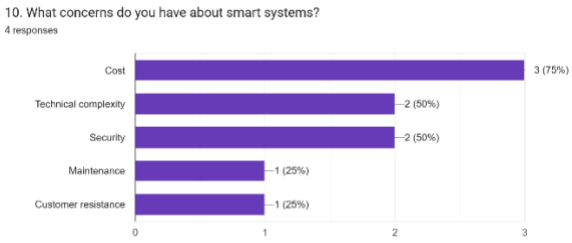



*Figure 4.6 Concerns of the owners*

Concerns about smart systems were also assessed. As shown in Figure 4.6, the most common concern among owners was cost, reported by 75 percent. Technical complexity and security concerns were each acknowledged by 50 percent of respondents. Maintenance requirements and customer resistance were identified by 25 percent. These findings show that while owners recognise the potential benefits of modern systems, they remain cautious about financial and technical implications. This highlights the importance of designing a solution that is affordable, simple to maintain and secure.

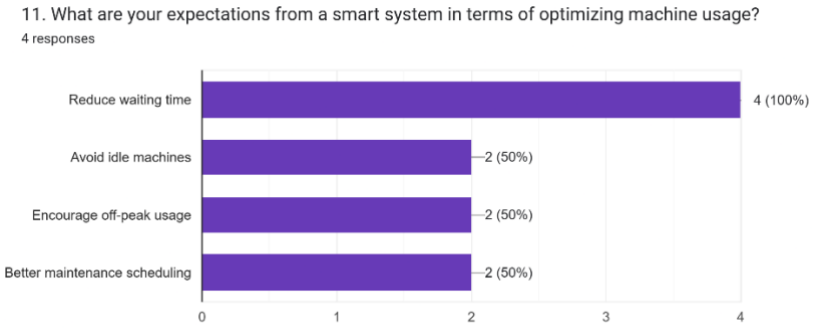



*Figure 4.7 Owners' expectations*

Finally, the survey examined what owners expect from a smart system in terms of improving machine usage. Figure 4.7 shows that all owners expect a system to help reduce waiting time. Half of the respondents noted that avoiding idle machines, encouraging off peak usage and improving maintenance scheduling would be valuable. These expectations align closely with the issues reported by users and reinforce the need for real time monitoring, automated usage tracking and data driven insights.

Overall, the preliminary survey confirms a clear gap between traditional laundromat operations and modern user expectations. Users experience uncertainty, long waiting periods

and a lack of transparency, while owners struggle with monitoring issues, machine misuse and limited data. The strong preference for real time updates, booking features and cloud accessibility validates the direction of this project. The findings directly informed the development of the Smart Laundromat Management System, ensuring that the proposed design addresses real operational pain points faced by both users and owners..

## 4.2 Acceptance Survey

The acceptance survey was conducted to evaluate the functionality and usability of the Smart Laundromat Management System after the prototype was completed. A total of twenty five respondents participated in the survey. The responses were used to assess how well the system met user needs and whether the IoT and cloud components performed reliably in supporting the intended features.

### 4.2.1 Respondent Profile

Figure 4.2.1 presents the distribution of respondents by role. Most respondents were laundromat users, representing 83.3 percent of the total sample, while 16.7 percent were laundromat business owners. This distribution is suitable for acceptance testing because it reflects both the customer side and the operational side of laundromat services. User responses focused mainly on the mobile features such as real time monitoring and booking, while owner responses assessed the practicality of the IoT module and the management dashboard.

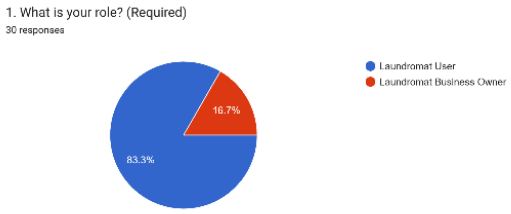



*Figure 4.2.1 Respondent Profile*

Respondents were also asked about the frequency of their laundromat visits. As shown in Figure 4.2.2, most respondents use laundromat services weekly, followed by a smaller group of monthly users. This pattern shows that the majority of respondents are regular laundromat users with consistent exposure to the typical challenges of machine availability, peak hour congestion and time planning. Their feedback is therefore valuable for evaluating the real world usefulness of the system.




*Figure 4.2.2 Visit Frequency*

### 4.2.2 System Functionality Evaluation

The acceptance survey included several questions related to system functionality. Figure 4.2.3 shows respondents' feedback regarding the accuracy of real time machine availability. Most respondents selected the highest rating, with a smaller portion giving a rating of four. This indicates high confidence in the performance of the IoT module and the cloud update process. The reliability of real time updates is an important achievement because it directly addresses the problem of uncertainty identified in the preliminary survey.

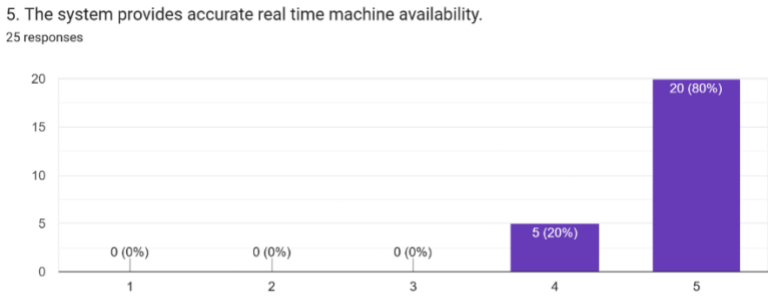



*Figure 4.2.3 Users' Feedback on System Functionality*

Responses related to cycle progress display, booking reliability and completion notifications also showed strong acceptance. Users agreed that the system helped reduce waiting times by allowing them to plan their visits more effectively. Many users noted that the notifications ensured they did not lose track of their laundry, especially when multitasking. These results show that the system's core functions were implemented successfully and performed reliably during testing.

### 4.2.3 Usability, Perceived Usefulness and User Experience

This combined section presents a detailed evaluation of the system's usability and its impact on user experience. The survey results show strong acceptance across all usability dimensions. Respondents reported that the mobile application was easy to navigate and that information was presented clearly. They agreed that the icons, colours and labels used in the system were intuitive, allowing them to understand machine states at a glance. These elements form the basis of a good user interface because clarity reduces cognitive effort and ensures that users can operate the system with minimal instruction.

Users also commented on the simplicity of the machine availability checking process. The transition between checking availability, viewing machine details and making a booking was described as smooth. The booking flow was appreciated for its logical sequence, which mirrors common reservation systems used in everyday applications. This indicates that the system successfully adopts familiar design principles, reducing the learning curve for new users.

Beyond usability, the acceptance responses show that the system improved the overall laundry experience. Many users mentioned that the real time status feature helped them avoid unnecessary trips, particularly during busy hours. The ability to view cycle progress from home or while doing other tasks allowed them to plan their time more effectively. Feedback also showed that users preferred laundromats that offer such technology because it increases transparency and convenience. The system's real time updates and booking function address the exact pain points that users identified during the preliminary survey, such as long waiting times and uncertainty about machine availability.

Taken together, the usability and usefulness results indicate that the system provides a practical and meaningful enhancement to the traditional laundromat experience. The interface supports quick decision making, while the system features reduce inconvenience and improve time management. These outcomes directly align with the project objectives and confirm the value of implementing IoT and cloud technologies in laundromat settings.

### 4.2.4 Business Owner Acceptance

Although business owners represented a smaller proportion of respondents, their feedback offers important insight into the operational benefits of the system. Figure 4.2.4 summarises owner responses to IoT performance and dashboard usability. Most owners agreed that the IoT module accurately detected machine states and updated the cloud database without significant delay. They indicated that the system logs, including timestamps and usage history, were complete and useful for understanding machine activity.

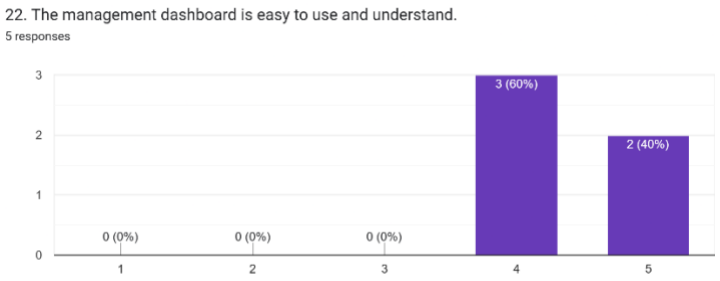



*Figure 4.2.4 Owners Feedback on dashboard feature*

Owners reported that the dashboard was easy to interpret, especially the machine status display and usage analytics. They highlighted that the system reduced the need for manual inspection because real time updates allowed them to monitor operations remotely. This finding is important because manual supervision is a major operational limitation in many laundromats, especially those without on-site staff.

Feedback also shows that owners see potential for the system to support maintenance planning. The ability to identify idle machines, underused machines or machines with irregular patterns can help detect faults early. Owners noted that usage analytics could improve scheduling and enable better business decisions, such as rearranging peak hour strategies or optimising machine placement.

Several owners also commented on feasibility. Most stated that the system is suitable for real deployment, although some expressed concerns about cost and long term maintenance. These

concerns are common in IoT adoption and provide direction for future improvement, such as simplifying sensor installation or reducing hardware costs.

Overall, the owner acceptance results demonstrate that the system is not only helpful for customers but also valuable for operational management. The findings support the idea that IoT and cloud-based monitoring can enhance machine oversight, reduce manual labour, and improve business efficiency.

## 4.3 Discussions

The findings from both the preliminary and acceptance surveys indicate that the Smart Laundromat Management System addressed the major limitations present in traditional laundromat settings. The preliminary survey highlighted recurring issues experienced by laundromat users, particularly the lack of real time machine information, uncertainty regarding machine availability and the absence of an organised booking mechanism. Users described situations where they arrived at the laundromat only to discover fully occupied machines and long waiting times. This problem was consistent among frequent users, especially those who visit weekly or monthly, which formed the majority in the preliminary dataset. These challenges justified the need for an automated and transparent management system.

The acceptance survey results show that the system effectively responded to these concerns. Respondents expressed high confidence in the accuracy of the real time availability feature, which forms the core of the system's functionality. Most users gave the highest rating for machine status accuracy, demonstrating that the IoT module performed reliably in updating the cloud database. The positive feedback suggests that the solution successfully fills the information gap that previously caused inconvenience, unnecessary trips and prolonged waiting times. Additionally, the booking function was widely accepted and appreciated for its clarity and smooth operation. This addresses the earlier concern that users often had no structured method to secure machine access, which sometimes led to disputes or repeated visits.

Usability feedback further supports the effectiveness of the system. Users stated that the application interface was easy to navigate and that icons and colour indicators were clear. These responses show that the system's design choices were suitable for users with varying levels of technological familiarity. The ability to check machine status, view cycle progress and receive notifications improved user control and reduced uncertainty, which aligns with the expectations recorded in the preliminary survey. The combined findings show that the system was not only functional but also meaningful in improving user experience.

Business owner feedback is equally important in evaluating the practicality of the system. The preliminary survey revealed several operational challenges, including machine misuse, peak time congestion, payment issues and customer disputes. Owners also expressed concern over the lack of real time monitoring tools and limited usage analytics. These issues result in inefficient operations and prevent informed decision making. The acceptance responses show that the IoT and dashboard components provided solutions to these problems. Owners agreed that the system detected machine states accurately and presented information in a clear and structured dashboard. They also highlighted the benefit of remote monitoring, which reduces reliance on manual checks, particularly for unmanned laundromats.

The ability of the system to provide complete logs and usage history addresses the earlier limitation of limited operational visibility. Owners recognised that machine usage analytics could support maintenance scheduling and help manage peak periods more effectively. Although some concerns about cost and long term maintenance were mentioned, most owners indicated that the system is suitable for real deployment and could improve business efficiency. This demonstrates that the system design meets the practical needs of small to medium sized laundromat operations.

Overall, the findings from both surveys confirm that the Smart Laundromat Management System achieved its intended objectives. The system addressed significant user challenges, enhanced transparency and introduced a more organised machine usage process. It also provided business owners with tools for operational oversight, data driven decision making and improved maintenance planning. The alignment between user expectations, owner requirements and system performance indicates that the proposed solution is valuable, functional and suitable for real world application. The results support the potential for wider adoption of IoT and cloud based solutions in modern laundromat environments.

## 4.4 System Interface

The system interface was developed to provide a clear, structured and user friendly experience for both laundromat users and business owners. The interface design follows simple navigation principles and uses recognisable icons, consistent colour schemes and intuitive layouts to ensure accessibility for users of different backgrounds. This section describes each main interface page and explains how the design supports the intended functionality of the Smart Laundromat Management System.

### 4.4.1 Home Page Interface

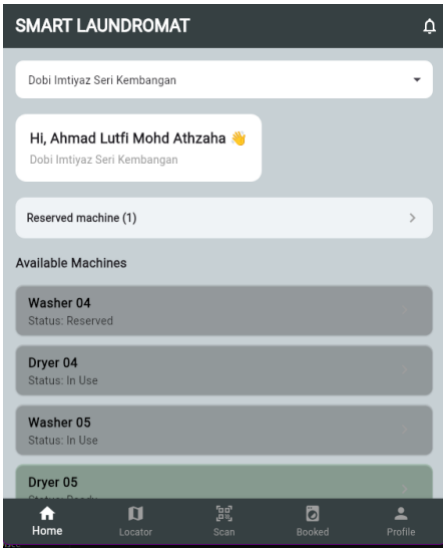



*Figure 4.4.1 Home Page Interface*

The Home Page serves as the central access point for users after successful login. It displays personalised information, machine status summaries and quick access to key features. The interface contains a greeting card that shows the user's name and selected store location. This feature was added to enhance user engagement and to confirm that the system has retrieved the correct account and store information from Firebase.

Below the greeting card, the Home Page presents a section that shows any machine reserved by the user. This section provides important visibility because users often forget which machine they booked, especially when managing multiple tasks. The clear display of reserved machines helps reduce confusion and improves time planning.

The largest component of the Home Page is the list of available and occupied machines. Machines are displayed with their names, such as “Washer 04” or “Dryer 05”, along with their real time status retrieved from the IoT module. The system uses colour coded cards to differentiate between machine conditions. Reserved machines are highlighted with a distinct colour, while those in use are shown in grey. Machines with the status “Ready” appear in green or a light tone. These colours allow users to identify machine states instantly without reading long descriptions.

This interface demonstrates the importance of real time updates. It addresses the concerns raised in the preliminary survey regarding machine availability uncertainty. The ability to view updated machine conditions supports efficient decision making and reduces unnecessary travel or waiting time.

### 4.4.2 Store Locator Interface

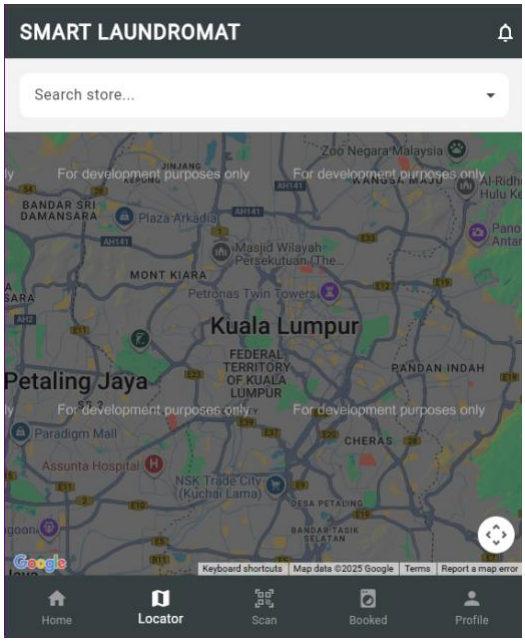



*Figure 4.4.2 Store Locator Interface*

The Store Locator page allows users to find nearby laundromats through an interactive map. The interface displays a search bar at the top, enabling users to search for a store by name or location. This is important for users who travel frequently or those who wish to compare machine availability across different branches.

The map is powered by Google Maps API and is embedded directly into the interface. The map shows store markers over Kuala Lumpur, Selangor and other surrounding regions. This view helps users identify the nearest branch based on their current location. It also aligns with the project objective of improving accessibility and providing a modern navigation feature that has become common in service-based applications.

The Store Locator interface addresses the issue highlighted in the preliminary survey where many users visit multiple laundromats before finding an available machine. By integrating a store map, the system makes it easier to choose a suitable branch before leaving home.

### 4.4.3 QR Scan Interface

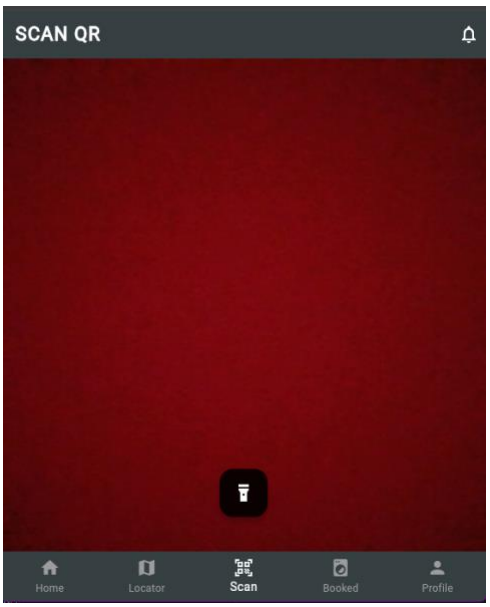



*Figure 4.4.3 QR Scan Interface*

The QR Scan page is designed to streamline the machine registration process. Instead of manually searching for a machine ID, users can scan a QR code placed on the washer or dryer. This page activates the phone camera and displays a scanning frame in the centre of the screen. The interface also includes a toggle flashlight button to assist scanning in low light conditions.

This functionality reduces user error and ensures that bookings or status checks correspond to the correct machine. QR code functionality is also aligned with modern service systems,

which makes it familiar to users who frequently use QR codes for payments or authentication.

The inclusion of a dedicated QR interface improves the system's accuracy and enhances user convenience, which contributes to a smoother overall experience.

### 4.4.4 Booked Machine Interface

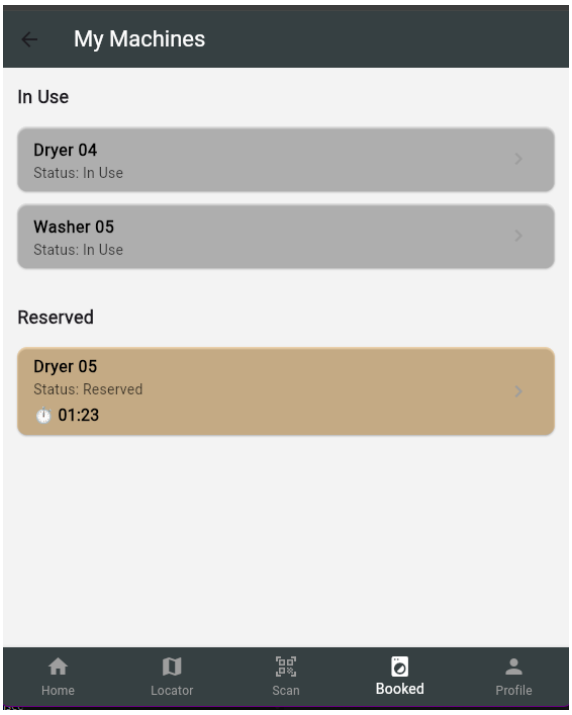



*Figure 4.4.4 Booked Machine Interface*

The Booked Machines page provides a complete list of machines that the user has either reserved or is currently operating. Machines are categorized into two sections. The first section titled "In Use" lists machines actively running under the user's account. The second section titled "Reserved" includes machines that the user has booked but not yet started.

This separation helps users manage their laundry activities more effectively. Each machine card shows real time status updates along with a countdown timer for reserved machines. The timer indicates how long the reservation will remain valid before the system automatically frees the machine.

The interface supports transparency and reduces disputes about machine ownership. It allows users to monitor their laundry remotely, which was one of the main issues highlighted in the

preliminary survey. The design ensures that users always have clear visibility of their machine usage and reservation status.

### 4.4.5 Profile Page Interface

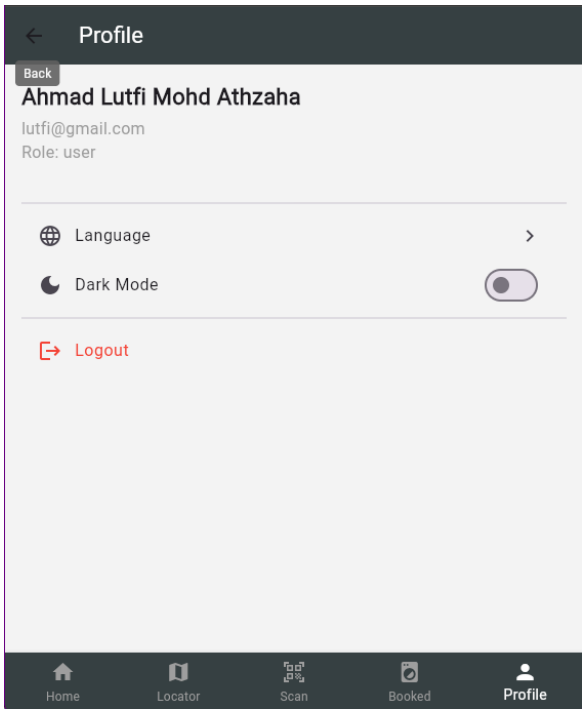



*Figure 4.4.5 Profile Page Interface*

The Profile page consolidates user personal information and account settings. The top portion displays the user's full name, email address and role. This confirms that the system has loaded the correct account details from Firebase, which is essential for security and user trust.

The Profile page includes options for changing language preferences and toggling between light mode and dark mode. These settings improve accessibility and allow users to adjust the interface according to their comfort. For example, dark mode reduces eye strain in low light environments.

A Logout button is provided at the bottom of the page. This function allows users to securely end their session. The Profile page reflects the system's commitment to user control and privacy.

### 4.4.6 Registration Page Interface

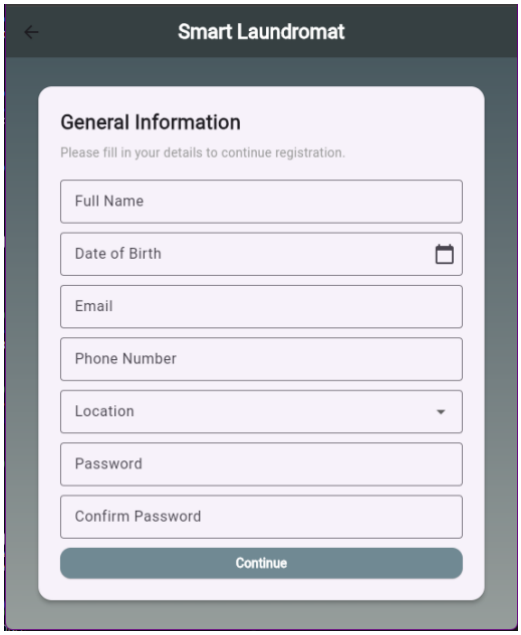



*Figure 4.4.6 Registration Page Interface*

The Registration page is the first user interaction with the application and plays an important role in establishing ease of use. The interface contains a structured form that captures essential information such as full name, date of birth, email, phone number, location and password credentials. Each input field is clearly labelled and supported with simple icons or drop-down menus.

The design ensures that users can complete registration without confusion. Field arrangement reflects a logical order, starting from personal details and ending with password creation. This design choice minimises errors and shortens the time required to complete registration.

The Registration page functionality is tied to Firebase Authentication, ensuring secure account creation. New users are guided through a process that verifies input completeness before continuing. This supports system reliability and reduces the likelihood of duplicated or incomplete records.

### 4.4.7 Interface Design Considerations

Across all pages, the system uses consistent colours, clear typography and intuitive button placements. These design choices reduce cognitive load and enhance the user experience. Navigation between the Home, Locator, Scan, Booked and Profile pages is supported by a bottom navigation bar that remains visible throughout the app. This ensures that users can always return to key functionalities with a single tap.

The interface design was heavily influenced by the findings from the preliminary survey. Users indicated that they struggled with transparency, waiting time and lack of structured booking systems. These concerns guided the integration of real time machine visibility, QR-based machine selection and a reservation management page. The layout and features directly address the limitations in traditional laundromats.

The interface also complements the IoT system by presenting sensor data in a clear format. Machine status updates appear immediately after the IoT module sends changes to the cloud database. This ensures accurate synchronisation between physical machines and the digital interface.

## 4.5 IoT Prototype Design and Operation

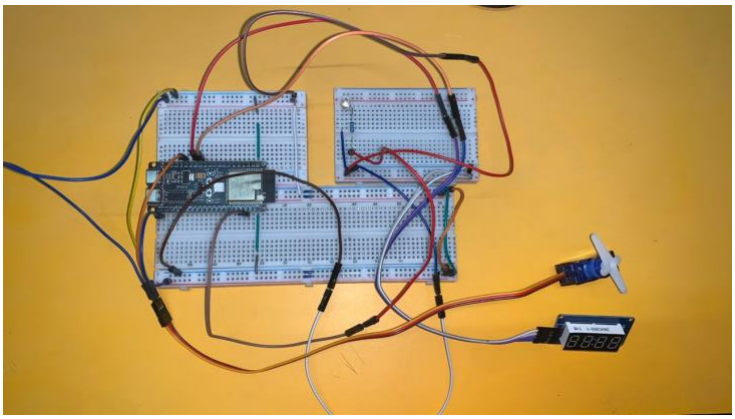


*Figure 4.5 IoT Prototype*

The IoT prototype functions as the hardware backbone of the Smart Laundromat Management System. It provides real time machine status detection, cycle monitoring and automated updates to the Firebase cloud database. The prototype integrates several electronic components mounted on interconnected breadboards, all controlled by an ESP32 microcontroller. The design was intentionally kept modular to support rapid testing, easy troubleshooting and clear separation of sensing, actuation and display functions.

Figure 4.x shows the assembled IoT prototype. The ESP32 microcontroller is positioned at the centre of the main breadboard. Supporting components are placed on adjacent breadboards and connected through jumper wires for power, signal transmission and grounding. The layout allows the system to simulate key laundromat machine behaviours such as machine activation, cycle duration measurement and remote locking through servo control.

The breadboard arrangement consists of:

- Primary breadboard with ESP32 microcontroller
- Secondary breadboard hosting the vibration or status indicator circuit
- Peripheral breadboard containing additional indicators such as the LED
- External components including the TM1637 display and SG90 servo motor

The prototype wiring follows a structured pattern where the ESP32 acts as the main processing unit, while each component receives dedicated signal pins and a shared power

supply.

The ESP32 microcontroller performs all processing tasks in the prototype. It reads sensor inputs, determines machine states and communicates updates to Firebase. The ESP32 was chosen because it supports built in WiFi, low power consumption and fast GPIO response. These characteristics make it ideal for IoT systems that require constant connectivity and reliable state detection.

In the prototype, the ESP32 executes several important tasks:

- Collects input from the machine status circuit (simulated vibration or load indicator)
- Triggers the servo motor for locking or unlocking machine lids
- Publishes real time machine state changes to Firebase
- Reads and displays countdown values on the TM1637 display
- Handles error detection such as loss of network or sensor unresponsiveness

The microcontroller bridges the gap between physical machine activity and the digital interface used by customers.

A secondary breadboard contains wires connected to an LED and sensor-input simulation circuit. This circuit represents the behaviour of a laundromat machine during a wash or dry cycle. When the simulated machine enters the “In Use” state, the ESP32 receives a high signal from the status wire. When the cycle ends, the circuit sends a low signal, indicating that the machine is idle or ready for new users.

This input is crucial because it drives the logic behind:

- Real time machine availability
- Booking validation
- Cycle progress timing
- Notifications triggered in the mobile app

In real deployment, this logic would connect to vibration sensors, current sensors or reed switches. For prototype purposes, the simpler circuit provides consistent and testable

behaviour without complex machine hardware.

The prototype uses a TM1637 four digit display to represent the cycle duration or countdown timer. This component mimics the digital timers commonly seen on commercial washing machines. The ESP32 sends the remaining cycle time to the display at regular intervals so the user can view real time progress.

The display performs two roles:

- Local feedback: It allows testing and debugging without needing to rely solely on the mobile app.

- Cycle event confirmation: When the countdown reaches zero, the ESP32 triggers a state change and updates Firebase, confirming the end of the machine cycle.

This improves internal timing accuracy and ensures the cloud database stays synchronised.

The SG90 micro servo motor is used to simulate the door locking mechanism of a washing machine. When a machine is in operation, the system should prevent users from opening the machine. The servo rotates to a closed position when the booked or active state is detected. Once the wash or dry cycle ends, the servo returns to an open position.

This demonstrates how the system can integrate physical control mechanisms into laundromat machines. In real-world applications, the servo could be replaced with a solenoid lock or relay controlled latch. The locking mechanism is also linked to the booking system. Once a reserved user activates the machine, the servo ensures exclusive access to prevent misuse or disputes.

A simple LED circuit is included to provide a visual indicator of machine state. When the machine is active, the LED lights up. When idle, the LED remains off. This helps during debugging and demonstrates real time responsiveness.

The LED feedback is also useful during testing to confirm:

- Correct transitions between machine states
- Proper synchronisation with Firebase updates
- Sensor signal accuracy

Even though this is a basic feature, it shows how machine feedback can be implemented with minimal hardware.

The wiring layout follows a clear separation between power and data lines. The ESP32 is powered through its USB connection, while the TM1637 and servo receive power from the breadboard rails. Red wires carry 5V or 3.3V supply depending on the component requirement, while black wires connect to ground.

Signal lines use distinct colours to differentiate functions such as:

- Data transmission from TM1637
- Pulse width modulation for the servo
- Status signal from the simulated machine
- Firebase communication via WiFi

Maintaining clean wiring improves the reliability of the prototype and reduces the chance of short circuits or unstable signals.

The ESP32 communicates with Firebase through WiFi. The communication flow is as follows:

- Sensor or simulated input sends machine state to ESP32
- ESP32 processes the input and updates machine status
- ESP32 sends a JSON payload to Firebase
- Firebase Firestore stores the updated status, time, and user ID
- Mobile application receives updates in real time through listeners

This architecture creates a complete closed loop between hardware and software. Every physical change in the prototype triggers an immediate update in the mobile interface, ensuring consistent communication between the user and the machine.

The IoT prototype demonstrates that physical machines can be digitised and synchronised with a cloud platform. It validates several important aspects of the project:

- Real time monitoring is technically feasible with low cost hardware

- Machine state transitions can be automated
- Cloud storage can reliably store and broadcast machine status
- The system design supports modular expansion for multiple machines
- Physical actuation (servo lock) can enhance fairness and safety in laundromat use

The prototype therefore plays a central role in confirming that the Smart Laundromat Management System is technically achievable and scalable.

## 4.6 Commercialisation Value and Research Contribution

### 4.6.1 Commercialisation Value

The system demonstrates strong commercial potential because it offers a low cost yet scalable solution. Many small laundromat businesses cannot afford high end commercial smart systems due to cost barriers. The prototype built in this project uses affordable sensors and open source cloud infrastructure, making it suitable for small enterprises.

The system also enhances user satisfaction, which can lead to higher customer retention. Real time availability updates and booking features help laundromats attract customers who prioritise convenience. Improved transparency and reduced waiting time can also increase the number of cycles per day, improving revenue.

The IoT dashboard allows operators to manage machines more effectively. This reduces maintenance delays and improves machine uptime. These factors increase overall operational efficiency and give small laundromats a competitive advantage.

### 4.6.2 Research Contribution

The project contributes to the research field by demonstrating how IoT and cloud technologies can be adapted for traditional service industries. It shows that low cost IoT components can provide accurate machine status and support real time systems.

The study also adds to the understanding of user acceptance in digitalised service environments. The results provide insight into how customers perceive smart booking, real time monitoring and cloud integrated mobile applications.

The system created in this project also serves as a foundation for future advancements such as predictive maintenance, machine learning based cycle optimisation and energy usage monitoring. These possibilities open pathways for further academic research and industry applications.

# CHAPTER 5

## CONCLUSION AND RECOMMENDATION

### 5.1 Conclusion

This project set out to design and implement a Smart Laundromat Management System using IoT and cloud technologies to address the long standing operational challenges in traditional self service laundromats. The preliminary findings showed that many users face uncertainty regarding machine availability, long waiting periods and a lack of real time information. Business owners expressed difficulty in monitoring machine usage, identifying operational inefficiencies and detecting machine faults without being physically present. These limitations highlighted the need for a digital system that improves transparency, organises customer flow and provides accurate operational data.

The prototype developed in this study demonstrates that IoT-based monitoring combined with Firebase cloud storage can provide reliable real time updates on machine status. The ESP32 microcontroller, servo locking mechanism and integrated timer display worked together to simulate actual laundromat machine behaviour. When paired with the mobile application, the system successfully provided features such as machine availability viewing, booking management, cycle tracking and notifications. User acceptance results indicated strong approval for the system's functionality and usability. Respondents reported that the system reduced uncertainty, improved time management and offered a more predictable laundry experience. They also expressed a preference for laundromats that implement similar smart systems.

From the business owner perspective, the system helped address key operational issues. Owners appreciated the real time dashboard, usage logs and monitoring capabilities. These functions enabled them to visualise peak hours, track machine performance and reduce reliance on manual inspection. Although some owners raised concerns related to cost and long term maintenance, most agreed that the system is suitable for real deployment, especially

for small and medium sized laundromat businesses seeking digital transformation.

Overall, the project achieved its objectives. The solution successfully demonstrated how IoT and cloud technologies can modernise laundromat operations, improve customer convenience and support better data-driven management. The combination of real time monitoring, booking, notifications and analytics forms a strong foundation for future enhancements. While the prototype works effectively at a conceptual level, there remains considerable potential for expansion and refinement. These opportunities form the basis of the recommendations for future researchers.

## 5.2 Recommendation

Future enhancements to the Smart Laundromat Management System can focus on transitioning the system from a reactive model to a more intelligent, predictive and automated platform. Based on the limitations observed during development and feedback from users and business owners, two key areas offer strong potential for future research.

### 5.2.1 Integration of Artificial Intelligence (AI) for Predictive Analytics

The current system provides real time information but does not provide predictions or insights about future machine usage. Future researchers can integrate AI or machine learning models to analyse historical machine data, user activity patterns and peak hour cycles to produce predictive analytics. This enhancement could allow the application to estimate waiting times before users arrive at the laundromat, predict when machines will become available and identify expected congestion periods.

For laundromat owners, AI can support data driven decision making. A predictive model can analyse long term usage to optimise maintenance schedules, helping prevent breakdowns by identifying early signs of wear or abnormal activity. It can also forecast busy periods, enabling owners to implement targeted promotions, adjust operational hours or redistribute resources across machines. Over time, AI could identify patterns such as repeated overuse of certain machines or unusual behaviour that may indicate misuse or the need for servicing.

In addition, predictive analytics can help build personalised recommendations for users,

guiding them to the best time to perform their laundry based on historical and real time data. This level of intelligence can significantly enhance convenience and improve overall customer satisfaction. Integrating AI would therefore elevate the system from a responsive tool to a smart and anticipatory solution.

### 5.2.2 Implementation of Image-Based Status Detection

The current IoT prototype relies on simulated sensor inputs to represent machine activity. Although this method is effective for testing, future researchers can explore the use of image-based monitoring through computer vision. A camera placed near the control panel or door area of a washing machine can capture visual cues to determine machine states such as running, idle or completed. This approach can reduce reliance on vibration or current sensors, making the system more flexible and easier to install across different machine models.

Image-based detection can offer several advantages. Computer vision systems can identify conditions that hardware sensors may not detect, such as improperly closed doors, overloaded drums or abandoned laundry after cycle completion. These situations often cause delays, errors or customer dissatisfaction, and addressing them automatically can improve operational efficiency. With appropriate image processing algorithms, the system can detect whether a machine door is blocked or not secured, preventing faulty starts or incomplete cycles.

Furthermore, integrating image analysis with deep learning models could enable the system to recognise machine faults or malfunction indicators displayed on the machine's panel. This capability would improve accuracy and reduce dependency on manual inspections. For laundromats with multiple machines, a camera-based solution can offer broader coverage and capture more detailed information about machine performance and user behaviour.

While implementing computer vision requires careful attention to privacy and data storage rules, it presents a powerful enhancement that future researchers can refine to create a more intelligent and autonomous monitoring platform.

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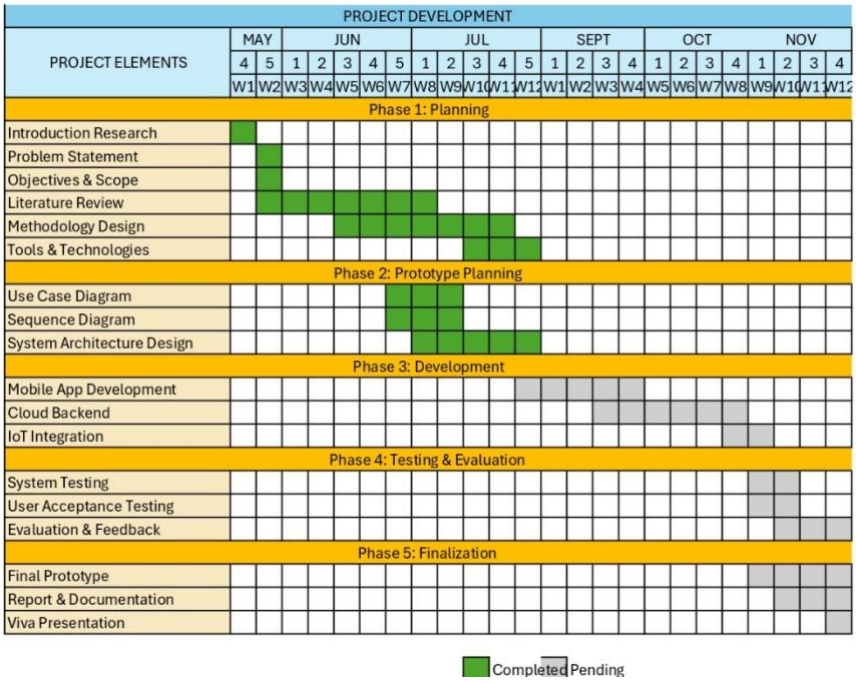
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# APPENDICES

Project Development Gantt Chart




*APPENDIX A Project Development Gantt Chart*